

# Type-specific catastrophe rates in a two-type branching process: an exact hypergeometric formula

## Abstract

For a continuous-time two-type birth–death process, we derive the exact finite-time probability that catastrophe has not yet occurred. Type 1 changes irreversibly into type 2, and an individual of either type can trigger a global absorbing catastrophe. Because the catastrophe rate depends on both populations along the path, the probability is an occupation-time transform. Its backward equations are triangular: the type-2 equation is autonomous, and substituting its solution into the type-1 equation leaves a Riccati equation. After linearisation, this becomes the Gauss hypergeometric equation. A Wronskian-normalised form keeps the coefficient enforcing the initial condition regular. The calculation also gives the probability of avoiding catastrophe forever and covers the zero-birth and zero-catastrophe-rate boundaries. With the appropriate initial data, the zero-catastrophe case recovers the two-type branching equations of Antal and Krapivsky. As one reading, the types may describe early and adapted intracellular phenotypes of *Yersinia pestis*, with catastrophe representing irreversible loss of macrophage containment. The formula is exact for any type-specific rate ordering, without an asymptotic or small-rate approximation.

## Contents

|          |                                                                      |           |
|----------|----------------------------------------------------------------------|-----------|
| <b>1</b> | <b>Introduction</b>                                                  | <b>2</b>  |
| <b>2</b> | <b>Model, assumptions, and backward equations</b>                    | <b>3</b>  |
| 2.1      | The two-type process . . . . .                                       | 3         |
| 2.2      | Occupation-time representation . . . . .                             | 5         |
| 2.3      | First-event calculation . . . . .                                    | 6         |
| 2.4      | Dimensionless form . . . . .                                         | 7         |
| <b>3</b> | <b>The autonomous type-2 probability</b>                             | <b>8</b>  |
| <b>4</b> | <b>Hypergeometric reduction and main result</b>                      | <b>11</b> |
| 4.1      | Selecting the stable constant part . . . . .                         | 11        |
| 4.2      | Linearising the Riccati equation . . . . .                           | 12        |
| 4.3      | The closed formula . . . . .                                         | 14        |
| 4.4      | Mathematical interpretation . . . . .                                | 18        |
| <b>5</b> | <b>Intracellular application: comparing catastrophe-rate regimes</b> | <b>18</b> |
| 5.1      | Intracellular mapping and scope . . . . .                            | 18        |
| 5.2      | Four catastrophe-rate regimes . . . . .                              | 19        |
| 5.3      | Exact comparison and limits of inference . . . . .                   | 20        |
| <b>6</b> | <b>Conclusion</b>                                                    | <b>21</b> |

|          |                                                                     |           |
|----------|---------------------------------------------------------------------|-----------|
| <b>A</b> | <b>From product of survival factors to the pathwise exponential</b> | <b>22</b> |
| <b>B</b> | <b>Hypergeometric derivatives and resonant parameters</b>           | <b>22</b> |
| <b>C</b> | <b>Checks, limits, and the earlier branching model</b>              | <b>22</b> |
| C.1      | Initial and infinite-time behaviour . . . . .                       | 23        |
| C.2      | Equal catastrophe rates . . . . .                                   | 23        |
| C.3      | Type 2 cannot trigger catastrophe . . . . .                         | 23        |
| C.4      | No type-1 births . . . . .                                          | 24        |
| C.5      | No type-2 births . . . . .                                          | 24        |
| C.6      | No conversion . . . . .                                             | 25        |
| <b>D</b> | <b>Equation-level recovery of Antal–Krapivsky</b>                   | <b>26</b> |
| <b>E</b> | <b>A catastrophe-aware population generating function</b>           | <b>26</b> |
| E.1      | The absorbing state and conditional populations . . . . .           | 27        |
| E.2      | Reuse of the hypergeometric solution . . . . .                      | 28        |
| E.3      | Three distributions encoded by the killed PGF . . . . .             | 29        |
| E.4      | Ultimate release counts . . . . .                                   | 30        |

# 1 Introduction

A population can disappear by ordinary deaths. It can also end for a different reason: one individual may trigger an event that stops the whole process. That event might be detection, intervention, or some other absorbing mechanism; the physical name depends on the application. The mathematical question is simpler: if individuals of two types contribute different amounts to the instantaneous catastrophe rate, what is the probability that the event has not happened by a finite time? The answer depends on how long each type is present, not only on the terminal counts, so that a lineage which grew early and then died can contribute more catastrophe exposure than a small lineage which remains alive. The natural object is therefore an average of exponential weights for total type-specific exposure: an occupation-time transform, with one weight for each type. Because any individual can trigger the same absorbing event, the model keeps a branching mechanism while the observable is global.

A concrete setting is the early intracellular phase of *Yersinia pestis* within macrophages. Under that reading, type 1 is an early intracellular state, conversion is progressive physiological adaptation rather than a genetic mutation, and type 2 is a later adapted state; catastrophe stands for irreversible loss of intracellular containment (macrophage burst, release of the bacterial load, or another terminal failure of the compartment), as distinct from ordinary bacterial death. The adapted phase could be more strongly associated with that endpoint, and unequal type-specific rates then let us compare that idea with alternatives; the biological background and the regimes are in [section 5](#). The exact avoidance probability simply asks how long containment can persist under any such mechanism.

The underlying population process is the familiar two-type birth–death model with one-way conversion. A type-1 individual can give birth, die, or change into type 2; a type-2 individual can give birth or die. Multitype branching processes provide the general setting [\[1\]](#). Antal and Krapivsky obtained an exact finite-time generating function for this triangular two-type process and used the hypergeometric structure created by its backward equations [\[2\]](#). Their calculation follows

terminal population counts; they obtain a pair of probability generating functions. Here we keep the same triangular structure and add type-specific routes to an absorbing catastrophe. The shared skeleton and the precise equation-level map to their system are set out in [appendix D](#); the short point is that their object and ours differ, so setting  $\delta_1 = \delta_2 = 0$  in the present initial-value problem does not recover their generating functions.

The extension gives the following two coupled nonlinear equations, derived in (3a) and (3b):

$$\begin{aligned}\frac{dS}{dt} &= \lambda_1 S^2 - (\lambda_1 + \mu_1 + \nu + \delta_1)S + \mu_1 + \nu G, \\ \frac{dG}{dt} &= \lambda_2 G^2 - (\lambda_2 + \mu_2 + \delta_2)G + \mu_2,\end{aligned}$$

with  $S(0) = G(0) = 1$ . The second equation is autonomous, which is why an exact calculation remains possible.

The derivation proceeds as follows.

- Solve the type-2 equation as a rational function of an exponentially decaying variable.
- Substitute that solution into the type-1 equation, leaving a non-autonomous Riccati equation.
- Shift by the long-time type-1 root, then linearise the Riccati equation by a logarithmic derivative.
- Change independent variable to the type-2 coordinate  $z_2$ , so that the linear equation becomes the Gauss hypergeometric equation.

The special function is forced by the reduction: it appears because the type-2 probability has one moving rational pole.

The principal result is a closed finite-time expression for the probability of avoiding catastrophe from one initial type-1 individual, with unequal type-specific catastrophe rates allowed. We also obtain the corresponding type-2 probability, identify both infinite-time limits, and treat the boundaries where either birth rate vanishes or type 2 cannot cause catastrophe. The calculation is exact within the model; what catastrophe *means* depends on the application. One reading, developed in [section 5](#), takes early and adapted intracellular phenotypes of *Y. pestis*, with containment failure as the absorbing event. The formulae themselves do not depend on that reading.

The paper is organised around the derivation. [Section 2](#) defines the process, gives its occupation-time representation, and derives the backward equations. [Section 3](#) solves the autonomous equation. The hypergeometric reduction and the main formula appear in [section 4](#), which closes with a short mathematical interpretation. [Section 5](#) then develops an intracellular application and compares catastrophe-rate regimes. Boundary cases and zero-birth formulae are collected in [appendix C](#); the map to Antal–Krapivsky is [appendix D](#). A short conclusion and four technical appendices complete the paper.

## 2 Model, assumptions, and backward equations

### 2.1 The two-type process

We begin with the process before catastrophe. Let  $X_t$  and  $Y_t$  denote the numbers of type-1 and type-2 individuals. A type-1 individual gives birth at rate  $\lambda_1$ , dies at rate  $\mu_1$ , and changes irreversibly into type 2 at rate  $\nu$ . A type-2 individual gives birth at rate  $\lambda_2$  and dies at rate  $\mu_2$ . All these rates are non-negative.

Catastrophe is a single global absorbing event. When the population is  $(x, y)$ , its instantaneous rate is

$$\delta_1 x + \delta_2 y, \quad (1)$$

where  $\delta_1, \delta_2 \geq 0$ . Thus one individual contributes a catastrophe rate determined by its type, but the first trigger sends the complete process to an absorbing state, denoted by  $(R, R)$ .

**Definition 2.1** (Birth–death–catastrophe process). Before absorption, the non-zero transition rates from  $(x, y) \in \mathbb{Z}_{\geq 0}^2$  are

|                                         |                                    |               |
|-----------------------------------------|------------------------------------|---------------|
| $(x, y) \longrightarrow (x + 1, y)$     | at rate $\lambda_1 x,$             | (birth)       |
| $(x, y) \longrightarrow (x - 1, y)$     | at rate $\mu_1 x,$                 | (death)       |
| $(x, y) \longrightarrow (x - 1, y + 1)$ | at rate $\nu x,$                   | (conversion)  |
| $(x, y) \longrightarrow (x, y + 1)$     | at rate $\lambda_2 y,$             | (birth)       |
| $(x, y) \longrightarrow (x, y - 1)$     | at rate $\mu_2 y,$                 | (death)       |
| $(x, y) \longrightarrow (R, R)$         | at rate $\delta_1 x + \delta_2 y.$ | (catastrophe) |

The state  $(R, R)$  is absorbing.

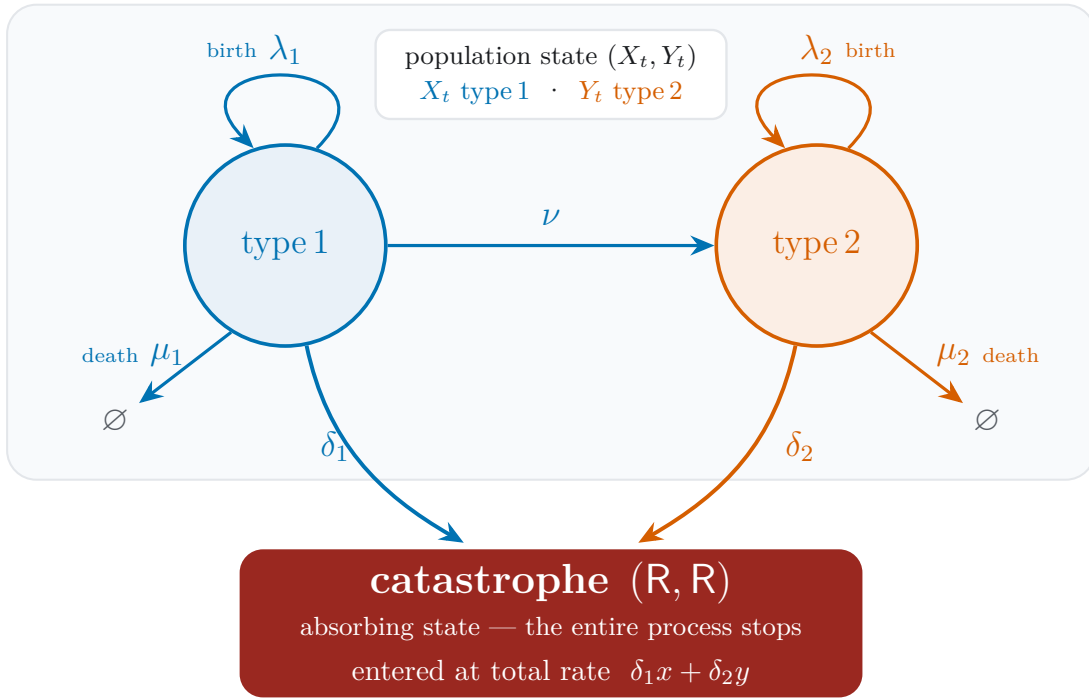


Figure 1: Two-type birth–death–conversion with a single global absorbing catastrophe. Type 1 (blue) gives birth at rate  $\lambda_1$ , dies at  $\mu_1$ , and converts irreversibly to type 2 (vermillion) at rate  $\nu$ ; type 2 gives birth at  $\lambda_2$  and dies at  $\mu_2$ . In state  $(x, y)$  catastrophe fires at rate  $\delta_1 x + \delta_2 y$  and absorbs the process in  $(R, R)$ .  $S(t)$  and  $G(t)$  are the probabilities that this has not occurred by time  $t$ , from one type-1 or type-2 individual respectively.

Write  $\tau_c$  for the catastrophe time. We need two probabilities because the initial type matters:

$$S(t) = \mathbb{P}_{(1,0)}\{\tau_c > t\},$$

$$G(t) = \mathbb{P}_{(0,1)}\{\tau_c > t\}.$$

They satisfy  $S(0) = G(0) = 1$ . These are no-catastrophe probabilities rather than population-survival probabilities: an ordinary death may remove the final individual without causing catastrophe, and such a path continues to count as catastrophe-free.

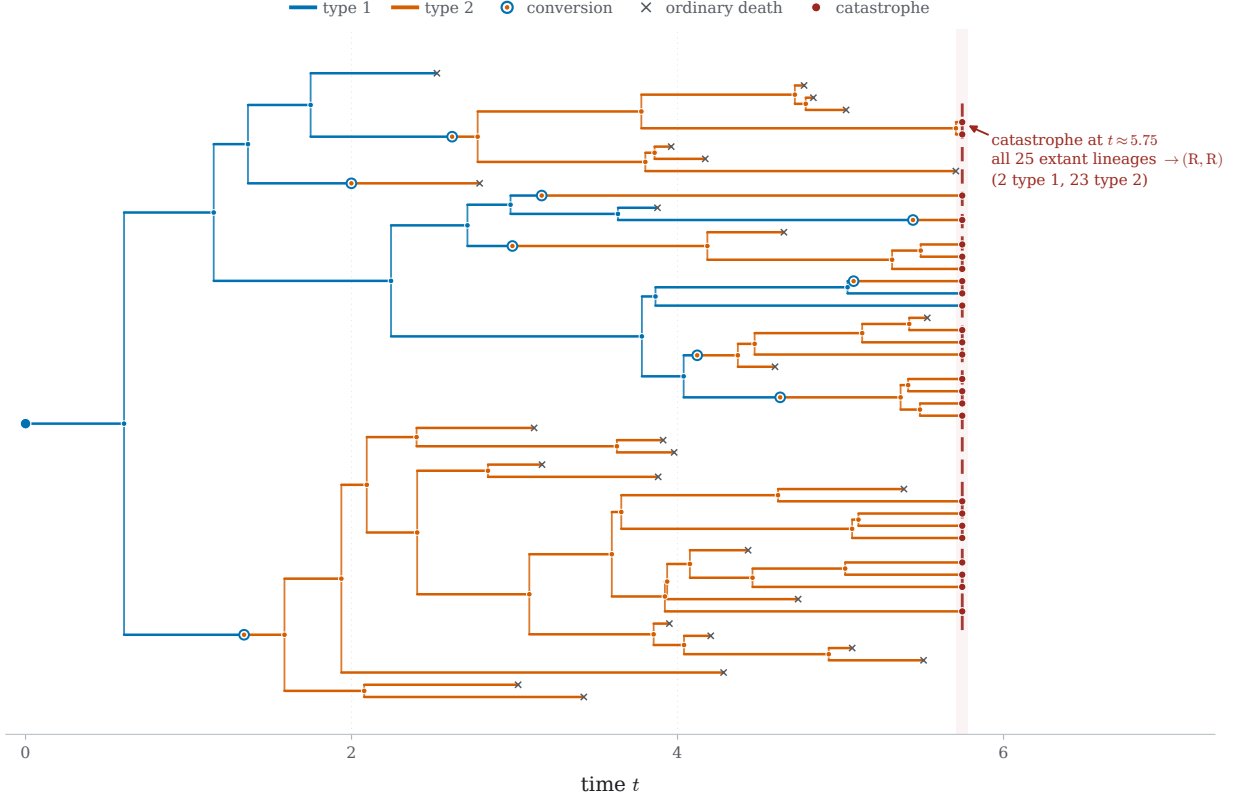


Figure 2: One continuous-time realisation starting from a single type-1 individual. Filled nodes mark births; blue-ringed vermilion nodes mark type 1  $\rightarrow$  2 conversion; crosses mark ordinary death; red discs mark lineages killed by the global catastrophe (dashed wavefront: simultaneous absorption in  $(R, R)$ ). Parameters:  $\lambda_1 = 1.0$ ,  $\mu_1 = 0.55$ ,  $\nu = 0.40$ ,  $\lambda_2 = 0.90$ ,  $\mu_2 = 0.50$ ,  $\delta_1 = 0.05$ ,  $\delta_2 = 0.08$  (Gillespie, seed 2869). [Genealogy visualiser](#)

## 2.2 Occupation-time representation

The same probabilities admit a second description that keeps ordinary extinction separate from catastrophe. Imagine first generating only the birth–death–conversion history  $(X_s, Y_s)_{0 \leq s \leq t}$ , as if the absorbing catastrophe jump were switched off. Each such history is one population path: a full record of how many type-1 and type-2 individuals are present at every time. Along that record, catastrophe is restored as a Poisson clock of rate  $\lambda(s) = \delta_1 X_s + \delta_2 Y_s$ . Over a short interval of length  $\Delta t$  the chance of no trigger is about  $1 - \lambda(s)\Delta t$ . The chance of no trigger on the whole of  $[0, t]$  is therefore a product of such factors, one per bin of a fine partition. As the mesh tends to zero that product becomes an exponential of a sum, and the sum becomes an integral (see [appendix A](#)):

$$\exp\left\{-\int_0^t \lambda(s) ds\right\} = \exp\left\{-\int_0^t (\delta_1 X_s + \delta_2 Y_s) ds\right\}.$$

Thus, given the path, this is the probability that the clock has not fired by time  $t$ . The expectation then averages that pathwise probability over random branching histories, recovering  $\mathbb{P}\{\tau_c > t\}$ .

**Proposition 2.2** (Occupation-time transform). *For the corresponding catastrophe-free branching process,*

$$S(t) = \mathbb{E}_{(1,0)} \left[ \exp \left\{ - \int_0^t (\delta_1 X_s + \delta_2 Y_s) ds \right\} \right], \quad (2a)$$

$$G(t) = \mathbb{E}_{(0,1)} \left[ \exp \left\{ - \int_0^t (\delta_1 X_s + \delta_2 Y_s) ds \right\} \right]. \quad (2b)$$

Consequently  $S$  and  $G$  lie in  $[0, 1]$  and are non-increasing in time.

*Proof.* Conditional on a path, (1) is the intensity of an inhomogeneous Poisson clock, whose survival probability is the displayed exponential. Taking the expectation over the branching path gives equation (2). The bounds and monotonicity hold pathwise, hence also after averaging.  $\square$

A lineage may die out by ordinary deaths and still contribute a positive weight to  $S(t)$  or  $G(t)$  if the catastrophe clock never rang. The calculation below uses backward equations, but this is the object those equations compute.

### 2.3 First-event calculation

The backward equations follow from the first event in a short interval  $\Delta t$ . Start with one type-1 individual. With high probability nothing happens and the no-catastrophe probability from that state remains  $S(t)$ . A birth produces two independent type-1 lineages, so the contribution is  $S(t)^2$ . An ordinary death leaves no one, and an empty population contributes probability one of avoiding future catastrophe. Conversion replaces the type-1 lineage by a type-2 lineage and contributes  $G(t)$ . Catastrophe in the interval means the event being measured has already failed, so that case contributes zero rather than a survival factor. The partition is summarised in figure 4. Collecting rates gives

$$\begin{aligned} S(t + \Delta t) &= [1 - (\lambda_1 + \mu_1 + \nu + \delta_1)\Delta t] S(t) \\ &\quad + \lambda_1 \Delta t S(t)^2 + \mu_1 \Delta t + \nu \Delta t G(t) + o(\Delta t). \end{aligned}$$

Starting from one type-2 individual, the same reasoning yields birth, death, inactivity, and catastrophe only (there is no conversion out of type 2), and we obtain

$$\begin{aligned} G(t + \Delta t) &= [1 - (\lambda_2 + \mu_2 + \delta_2)\Delta t] G(t) \\ &\quad + \lambda_2 \Delta t G(t)^2 + \mu_2 \Delta t + o(\Delta t). \end{aligned}$$

Subtract the value at  $t$ , divide by  $\Delta t$ , and let  $\Delta t$  tend to zero. We obtain the triangular backward system

$$\frac{dS}{dt} = \lambda_1 S^2 - (\lambda_1 + \mu_1 + \nu + \delta_1)S + \mu_1 + \nu G, \quad (3a)$$

$$\frac{dG}{dt} = \lambda_2 G^2 - (\lambda_2 + \mu_2 + \delta_2)G + \mu_2, \quad (3b)$$

with  $S(0) = G(0) = 1$ . Immediately,

$$S'(0) = -\delta_1, \quad G'(0) = -\delta_2. \quad (4)$$

The equations therefore reproduce the initial catastrophe rates before any birth, death, or conversion has had time to occur.

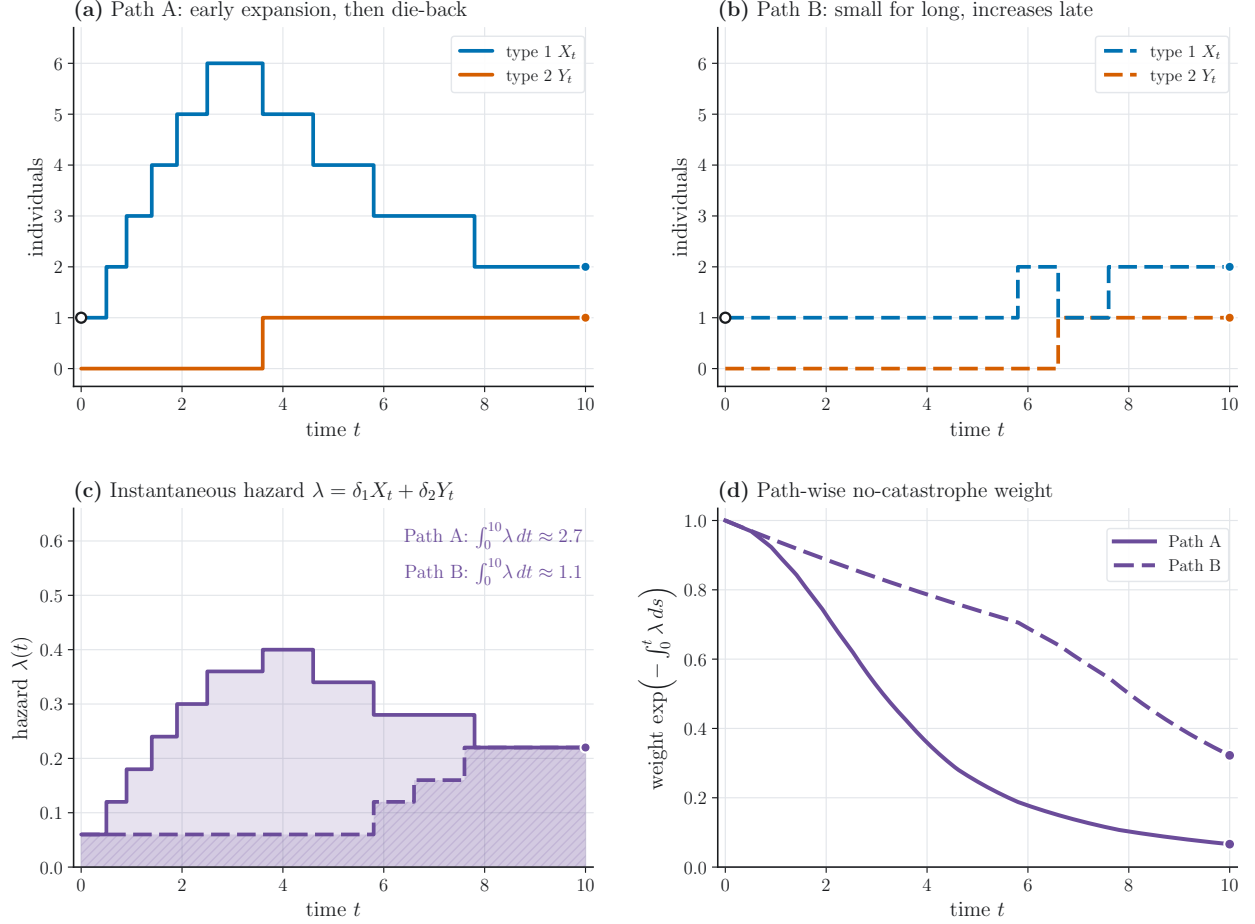


Figure 3: Why the no-catastrophe probability depends on occupation times, not only terminal counts. Two deterministic histories on  $t \in [0, 10]$  share  $(X_0, Y_0) = (1, 0)$  and  $(X_{10}, Y_{10}) = (2, 1)$ : (a) early peak then die-back; (b) late bloom. (c) Rates  $\lambda = \delta_1 X + \delta_2 Y$  share the terminal hazard but differ in integrated exposure by about 2.4. (d) Pathwise weights  $\exp\{-\int \lambda ds\}$  averaged in proposition 2.2 ( $\delta_1 = 0.15$ ,  $\delta_2 = 0.10$ ).

## 2.4 Dimensionless form

It is convenient to measure time in units of the type-1 birth timescale. We therefore assume from this point that

$$\lambda_1 > 0, \quad (5)$$

and set  $\hat{t} = \lambda_1 t$ . Define

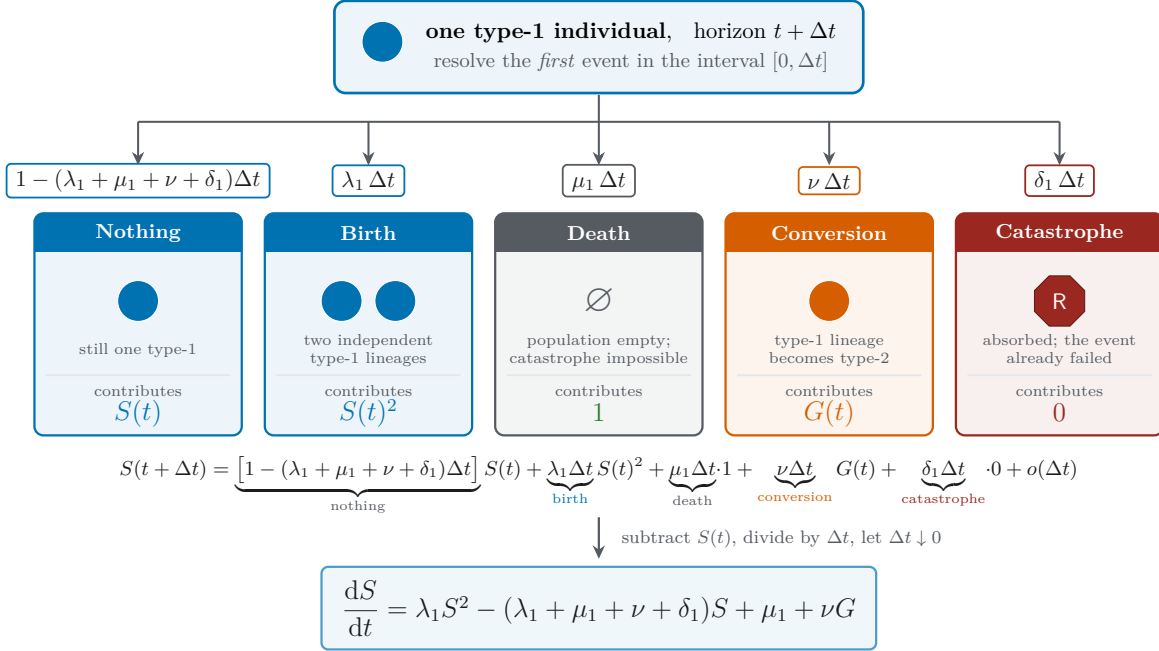
$$\hat{S}(\hat{t}) = S(\hat{t}/\lambda_1), \quad \hat{G}(\hat{t}) = G(\hat{t}/\lambda_1),$$

together with

$$\hat{\mu}_i = \frac{\mu_i}{\lambda_1}, \quad \hat{\nu} = \frac{\nu}{\lambda_1}, \quad \hat{\delta}_i = \frac{\delta_i}{\lambda_1}, \quad \hat{\lambda}_2 = \frac{\lambda_2}{\lambda_1}. \quad (6)$$

Every rate is now a dimensionless ratio to  $\lambda_1$ , and the type-1 birth coefficient in the scaled equations is one. With primes denoting differentiation in  $\hat{t}$ , the system becomes

$$\hat{S}' = \hat{S}^2 - \hat{\kappa}_1 \hat{S} + \hat{\mu}_1 + \hat{\nu} \hat{G}, \quad \hat{\kappa}_1 = 1 + \hat{\mu}_1 + \hat{\nu} + \hat{\delta}_1, \quad (7a)$$



$$\hat{G}' = \hat{\lambda}_2 \hat{G}^2 - (\hat{\lambda}_2 + \hat{\mu}_2 + \hat{\delta}_2) \hat{G} + \hat{\mu}_2. \quad (7b)$$

For the main formula we take  $\hat{\lambda}_2 > 0$ ,  $\hat{\delta}_2 > 0$ , and  $\hat{\nu} > 0$ . The cases  $\lambda_1 = 0$ ,  $\hat{\lambda}_2 = 0$ ,  $\hat{\delta}_2 = 0$ , and  $\hat{\nu} = 0$  are separated in [appendix C](#). Those boundaries matter: the scaling or the coordinate used in the hypergeometric reduction degenerates there, even though the original unscaled initial-value problem remains regular.

### 3 The autonomous type-2 probability

The second equation in (7) does not depend on  $\hat{S}$ : it is autonomous. This is the small structural fact on which the full calculation rests. We solve it first: factor the right-hand side, introduce a decaying coordinate  $z_2$ , and write  $\hat{G}$  as a rational function of  $z_2$ . That rational time dependence will then drive the type-1 equation.

Constant solutions of [equation \(7b\)](#) satisfy a quadratic equation. Setting

$$\hat{\Delta}_2 = \sqrt{(\hat{\lambda}_2 + \hat{\mu}_2 + \hat{\delta}_2)^2 - 4\hat{\lambda}_2\hat{\mu}_2}, \quad (8)$$

the two roots are

$$r_{2,\pm} = \frac{\hat{\lambda}_2 + \hat{\mu}_2 + \hat{\delta}_2 \pm \hat{\Delta}_2}{2\hat{\lambda}_2}. \quad (9)$$

They satisfy

$$r_{2,-} + r_{2,+} = \frac{\hat{\lambda}_2 + \hat{\mu}_2 + \hat{\delta}_2}{\hat{\lambda}_2}, \quad r_{2,-} r_{2,+} = \frac{\hat{\mu}_2}{\hat{\lambda}_2},$$

and therefore the right-hand side factors as

$$\hat{G}' = \hat{\lambda}_2 (\hat{G} - r_{2,-})(\hat{G} - r_{2,+}). \quad (10)$$

There is also a probabilistic ordering among these equilibria. The quadratic is non-negative at zero and equals  $-\hat{\delta}_2 < 0$  at one, while its leading coefficient is positive. Therefore

$$0 \leq r_{2,-} < 1 < r_{2,+}.$$

Note that a long-time probability must lie in  $[0, 1]$ , so only the lower root  $r_{2,-}$  can be an admissible attractor; the upper root lies outside that interval. Figure 5 makes this geometry and the subsequent relaxation explicit.

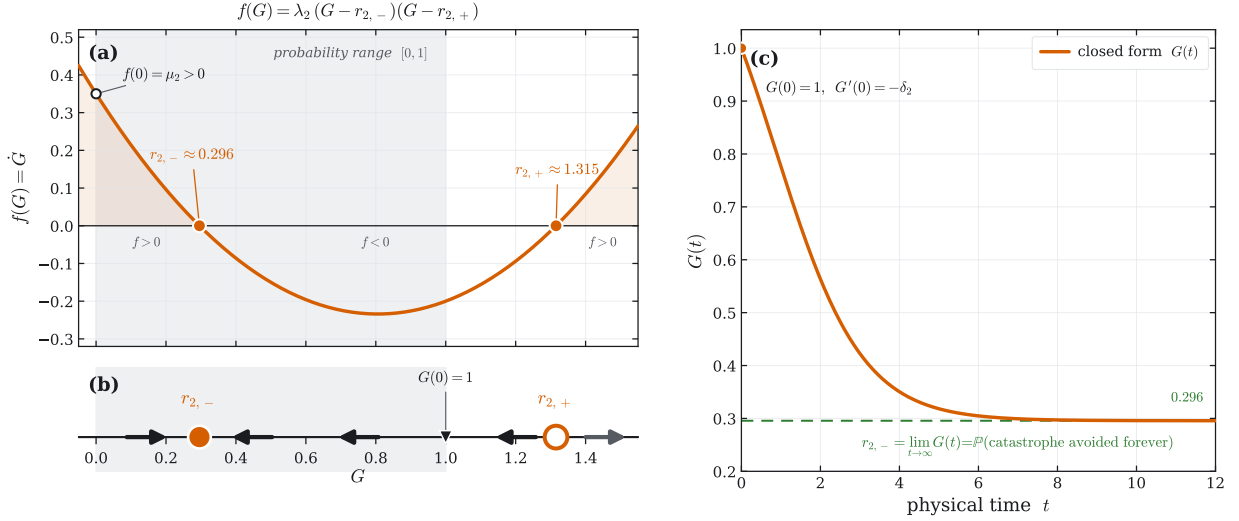


Figure 5: Geometry of the autonomous type-2 probability. (a) Right-hand side  $f(G) = \lambda_2(G - r_{2,-})(G - r_{2,+})$ , with  $0 \leq r_{2,-} < 1 < r_{2,+}$  so only  $r_{2,-}$  is an admissible long-time probability. (b) Phase line:  $r_{2,-}$  attracting,  $r_{2,+}$  repelling;  $G(0) = 1$  is drawn into  $r_{2,-}$ . (c) Relaxation of  $G(t)$  from 1 toward  $r_{2,-}$ . Parameters:  $\lambda_2 = 0.9$ ,  $\mu_2 = 0.35$ ,  $\delta_2 = 0.20$ .

We shall use the root gap and its associated decay rate repeatedly. Define

$$\alpha_2 = r_{2,+} - r_{2,-} = \frac{\hat{\Delta}_2}{\hat{\lambda}_2}, \quad \hat{q}_2 = -\hat{\lambda}_2 \alpha_2 = -\hat{\Delta}_2. \quad (11)$$

Thus  $\alpha_2 > 0$  and  $\hat{q}_2 < 0$ .

**Proposition 3.1** (Exact type-2 probability). *Let*

$$z_{2,0} = \frac{r_{2,-} - 1}{r_{2,+} - 1} < 0, \quad z_2(\hat{t}) = z_{2,0} e^{\hat{q}_2 \hat{t}}. \quad (12)$$

*Then the solution of equation (7b) with  $\hat{G}(0) = 1$  is*

$$\hat{G}(\hat{t}) = r_{2,-} - \alpha_2 \frac{z_2(\hat{t})}{1 - z_2(\hat{t})}. \quad (13)$$

*In physical time, with  $\hat{t} = \lambda_1 t$  and the scaled rates of equation (6),*

$$G(t) = r_{2,-} - \alpha_2 \frac{z_2(\lambda_1 t)}{1 - z_2(\lambda_1 t)}.$$

*Proof.* Separate equation (10) and use partial fractions:

$$\frac{d\hat{G}}{(\hat{G} - r_{2,-})(\hat{G} - r_{2,+})} = \hat{\lambda}_2 d\hat{t}.$$

After integration,

$$\log\left(\frac{\hat{G} - r_{2,-}}{\hat{G} - r_{2,+}}\right) = -\hat{\lambda}_2 \alpha_2 \hat{t} + \text{constant}.$$

The left-hand ratio is the natural coordinate produced by separation of variables. The initial condition makes it equal to  $z_{2,0}$  at zero, so it equals  $z_2(\hat{t})$  at time  $\hat{t}$ . Solving the resulting linear fractional equation for  $\hat{G}$  gives equation (13).  $\square$

Thus  $z_2$  is simply that separated ratio, started from  $\hat{G}(0) = 1$  and decaying exponentially toward zero as  $\hat{G}$  approaches  $r_{2,-}$ . Three immediate checks are

$$\hat{G}(0) = 1, \quad \hat{G}'(0) = -\hat{\delta}_2, \quad \hat{G}(\hat{t}) \rightarrow r_{2,-} \quad (\hat{t} \rightarrow \infty).$$

The last limit has a direct meaning. Starting from one type-2 individual,  $r_{2,-}$  is the probability that catastrophe is avoided forever. It can be positive because the lineage may die out ordinarily before it triggers anything.

An expanded form may be useful for direct implementation:

$$\hat{G}(\hat{t}) = \frac{r_{2,-}(r_{2,+} - 1) + r_{2,+}(1 - r_{2,-})e^{-\hat{\lambda}_2(r_{2,+} - r_{2,-})\hat{t}}}{(r_{2,+} - 1) + (1 - r_{2,-})e^{-\hat{\lambda}_2(r_{2,+} - r_{2,-})\hat{t}}}.$$

In physical time the same expression is  $G(t) = \hat{G}(\lambda_1 t)$ . The compact coordinate  $z_2$  is the better choice for the driven equation: it is always negative, moves monotonically toward zero, and carries all of the time dependence of  $\hat{G}$  in a rational way (figure 6). With  $\hat{G}$  known, the type-1 equation is now a forced Riccati equation in  $\hat{S}$ . In the next section we substitute this  $G$  and use  $z_2$  as the independent variable.

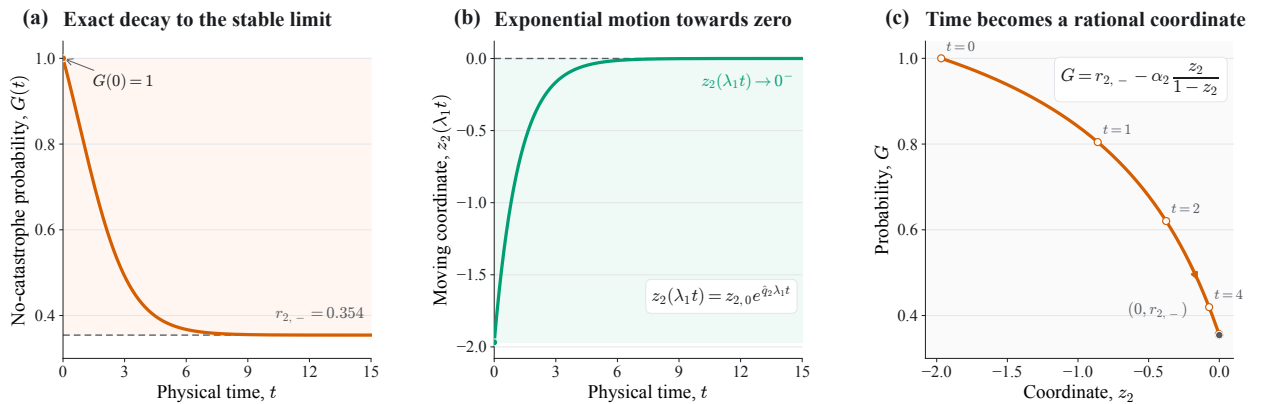
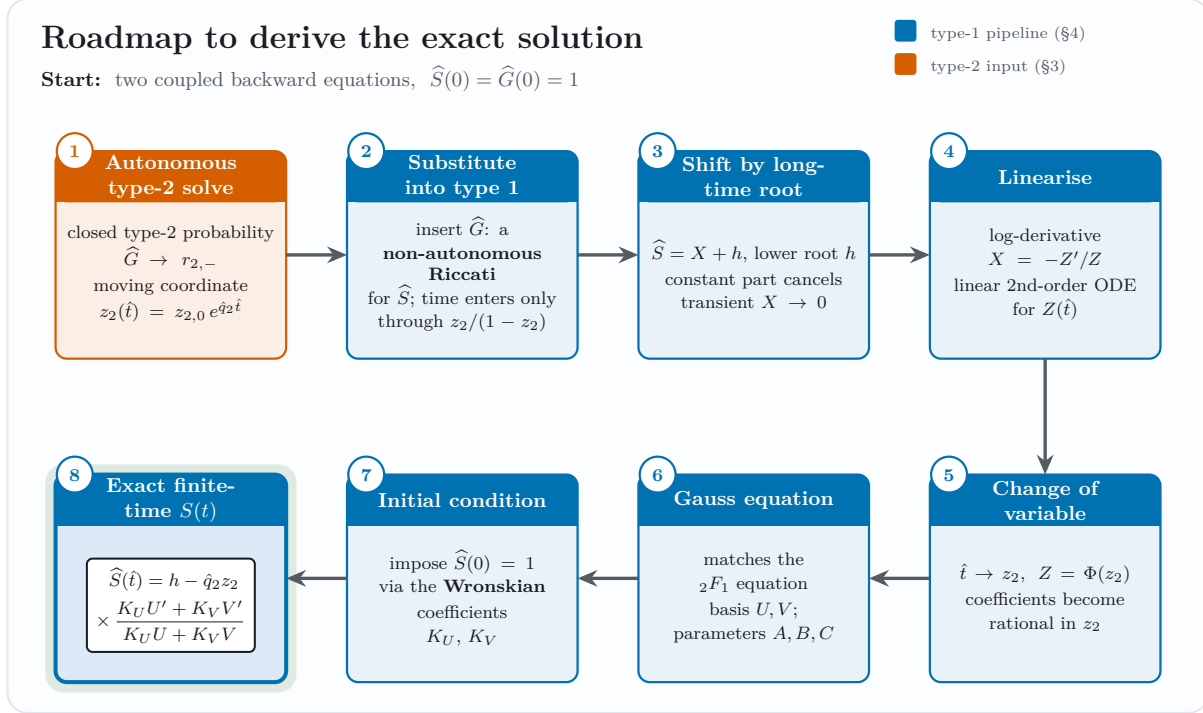


Figure 6: (a)  $G(t)$  from 1 toward  $r_{2,-}$ . (b) Coordinate  $z_2(\lambda_1 t)$  increasing from  $z_{2,0} < 0$  toward 0. (c) Linear-fractional map  $G = r_{2,-} - \alpha_2 z_2 / (1 - z_2)$ . Parameters:  $\lambda_1 = 1$ ,  $\lambda_2 = 0.85$ ,  $\mu_2 = 0.40$ ,  $\delta_2 = 0.18$ .

## 4 Hypergeometric reduction and main result

With the type-2 solution in hand, we substitute [equation \(13\)](#) into the equation for  $\hat{S}$ . The result is a forced Riccati equation whose time dependence occurs only through  $z_2/(1-z_2)$ . We shift by a long-time root, linearise with the standard Riccati trick, change independent variable to  $z_2$ , match the Gauss hypergeometric equation, and fix the free mix of solutions by  $\hat{S}(0) = 1$ . The method is inspired by that of Antal and Krapivsky; the subsections below present each of the steps.



### 4.1 Selecting the stable constant part

As  $\hat{t} \rightarrow \infty$ , we have  $\hat{G} \rightarrow r_{2,-}$ . With type 2 settled, the equation for  $\hat{S}$  becomes autonomous, so any long-time limit of  $\hat{S}$  must be a constant solution of the algebraic equation obtained by setting  $\hat{S}' = 0$  with  $\hat{G} = r_{2,-}$ . That constant is a long-time root of the limiting type-1 equation. Write

$$\hat{S}(\hat{t}) = X(\hat{t}) + h,$$

where  $h$  is a constant still to be set and  $X$  is the remaining time-dependent deviation from that plateau. Direct substitution into [equation \(7a\)](#) gives

$$\begin{aligned} X' = X^2 + (2h - \hat{\kappa}_1)X - \hat{\nu}\alpha_2 \frac{z_2}{1-z_2} \\ + [h^2 - \hat{\kappa}_1 h + \hat{\mu}_1 + \hat{\nu}r_{2,-}]. \end{aligned}$$

We choose  $h$  so that the final bracket vanishes. The resulting quadratic is

$$h^2 - \hat{\kappa}_1 h + \hat{\mu}_1 + \hat{\nu}r_{2,-} = 0, \tag{14}$$

and we take the lower root

$$h = \frac{\hat{\kappa}_1 - \sqrt{\hat{\kappa}_1^2 - 4(\hat{\mu}_1 + \hat{\nu}r_{2,-})}}{2}. \quad (15)$$

With that choice,  $h$  is exactly the long-time root: once  $X \rightarrow 0$ , one has  $\hat{S} \rightarrow h$ , which is the forever-avoidance probability from one initial type-1 individual. The coordinate  $z_2$  is not needed to define  $h$ ; only the settled value  $r_{2,-}$  of  $\hat{G}$  enters [equation \(14\)](#).

We take the lower root because it is the stable attractor inside the admissible probability interval. Since  $r_{2,-} \leq 1$ , the discriminant is bounded below by

$$\begin{aligned} \hat{\kappa}_1^2 - 4(\hat{\mu}_1 + \hat{\nu}r_{2,-}) &\geq (1 + \hat{\mu}_1 + \hat{\nu} + \hat{\delta}_1)^2 - 4(\hat{\mu}_1 + \hat{\nu}) \\ &= (u - 1)^2 + 2\hat{\delta}_1(1 + u) + \hat{\delta}_1^2, \end{aligned}$$

where  $u = \hat{\mu}_1 + \hat{\nu}$ , so the roots are real. Moreover

$$2h - \hat{\kappa}_1 = -\sqrt{\hat{\kappa}_1^2 - 4(\hat{\mu}_1 + \hat{\nu}r_{2,-})} < 0,$$

so the lower root is the stable attractor of the limiting autonomous equation, while the other root lies above one in the genuinely coupled catastrophe regime. With this choice,  $X = \hat{S} - h$  satisfies

$$X' = X^2 + (2h - \hat{\kappa}_1)X - \hat{\nu}\alpha_2 \frac{z_2}{1 - z_2}. \quad (16)$$

## 4.2 Linearising the Riccati equation

We now want a linear equation. A Riccati equation of the form  $X' = X^2 + \dots$  linearises under the standard Riccati trick  $X = -(\log Z)'$  [\[3\]](#). Set

$$X = -\frac{Z'}{Z}. \quad (17)$$

Then

$$X' = -\frac{Z''}{Z} + \left(\frac{Z'}{Z}\right)^2 = -\frac{Z''}{Z} + X^2.$$

The quadratic terms cancel in [equation \(16\)](#), and we land on the linear second-order equation

$$Z'' + (\hat{\kappa}_1 - 2h)Z' - \hat{\nu}\alpha_2 \frac{z_2(\hat{t})}{1 - z_2(\hat{t})}Z = 0, \quad (18)$$

in which the only remaining awkwardness is the time-dependent coefficient of  $Z$ . Because  $X$  depends on  $Z$  only through the logarithmic derivative  $Z'/Z$ , any overall multiplicative constant in  $Z$  cancels from the probability.

### 4.2.1 Change of independent variable

All remaining time dependence in [equation \(18\)](#) sits inside  $z_2(\hat{t})$ : the coefficient is  $z_2/(1 - z_2)$ , and  $z_2$  itself is a known exponential function of  $\hat{t}$ . Measuring “time” by  $z_2$  rather than by  $\hat{t}$  turns a linear ODE with exponential-looking coefficients into the Gauss hypergeometric equation, whose solutions are known.

Since  $z_2' = \hat{q}_2 z_2$  and  $z_2$  is monotone, write

$$Z(\hat{t}) = \Phi(z_2(\hat{t})).$$

If primes on  $\Phi$  mean differentiation with respect to  $z_2$ , the chain rule gives

$$Z' = \hat{q}_2 z_2 \Phi', \quad (19)$$

and then

$$Z'' = \hat{q}_2^2 (z_2^2 \Phi'' + z_2 \Phi'). \quad (20)$$

Substitute [equations \(19\) and \(20\)](#) into [equation \(18\)](#). For every finite  $\hat{t}$ ,  $z_2 \neq 0$ , so we may divide by  $\hat{q}_2^2 z_2$  and obtain the intermediate form

$$z_2 \Phi'' + \left(1 + \frac{\hat{\kappa}_1 - 2h}{\hat{q}_2}\right) \Phi' - \frac{\hat{\nu} \alpha_2}{\hat{q}_2^2} \frac{1}{1 - z_2} \Phi = 0.$$

Multiplying through by  $(1 - z_2)$  clears the remaining denominator and yields

$$z_2(1 - z_2)\Phi'' + C(1 - z_2)\Phi' - \frac{\hat{\nu} \alpha_2}{\hat{q}_2^2} \Phi = 0, \quad (21)$$

where

$$C = 1 + \frac{\hat{\kappa}_1 - 2h}{\hat{q}_2}. \quad (22)$$

Because  $\hat{q}_2 = -\hat{\Delta}_2$  and  $\hat{\kappa}_1 - 2h > 0$ , this parameter always satisfies

$$C = 1 - \frac{\hat{\kappa}_1 - 2h}{\hat{\Delta}_2} < 1.$$

#### 4.2.2 Matching to the Gauss equation

Equation (21) is an instance of the canonical Gauss hypergeometric equation

$$z(1 - z)\Phi'' + [C - (A + B + 1)z]\Phi' - AB\Phi = 0.$$

When  $C \notin \mathbb{Z}$ , its standard fundamental solutions are

$$\Phi_1(z) = {}_2F_1(A, B; C; z), \quad (23a)$$

$$\Phi_2(z) = z^{1-C} {}_2F_1(A - C + 1, B - C + 1; 2 - C; z). \quad (23b)$$

Every solution is then a linear combination  $\Phi = c_1 \Phi_1 + c_2 \Phi_2$ . The parameters  $A$ ,  $B$ , and  $C$  are fixed by matching coefficients with [equation \(21\)](#). Comparing the coefficient of  $\Phi'$  already gave  $C$  in [equation \(22\)](#). The remaining matches are

$$A + B = C - 1 = \frac{\hat{\kappa}_1 - 2h}{\hat{q}_2}, \quad AB = \frac{\hat{\nu} \alpha_2}{\hat{q}_2^2}. \quad (24)$$

Thus  $A$  and  $B$  are the two roots of one quadratic:

$$A, B = \frac{C - 1}{2} \pm \sqrt{\frac{(C - 1)^2}{4} - \frac{\hat{\nu} \alpha_2}{\hat{q}_2^2}}. \quad (25)$$

They need not be individually real. If the discriminant is negative, they form a complex-conjugate pair; the differential equation and its physical boundary data remain real. At this point the differential equation for  $\Phi$  is completely specified. What remains is to rebuild the probability and fix the free mix of the two solutions.

### 4.3 The closed formula

#### 4.3.1 Rebuilding $\hat{S}$ from $\Phi$

We first reverse the substitutions, before choosing a basis for  $\Phi$ . From

$$\hat{S} = h + X, \quad X = -\frac{Z'}{Z}, \quad Z = \Phi(z_2),$$

the chain rule [equation \(19\)](#) gives

$$\hat{S}(\hat{t}) = h - \hat{q}_2 z_2(\hat{t}) \frac{\Phi'(z_2(\hat{t}))}{\Phi(z_2(\hat{t}))}. \quad (26)$$

The probability depends on  $\Phi$  only through the logarithmic derivative  $\Phi'/\Phi$ . Any overall multiplicative constant in  $\Phi$  cancels, as already noted for  $Z$ .

#### 4.3.2 Ansatz for $\Phi$ : the two-dimensional solution space

Equation (21) is a second-order linear ODE, so its solution space is two-dimensional. We do not guess  $\Phi$  as a special function of the probability; we take the general solution of the equation we have derived. For  $C \notin \mathbb{Z}$  that general solution is any mix of the standard pair [equation \(23\)](#).

Our path has  $z_2(\hat{t}) < 0$  for all finite  $\hat{t}$ , so we evaluate everything on the negative real axis. The textbook second solution uses the factor  $z^{1-C}$ , which need not stay real for  $z < 0$  and real non-integer  $C$ . On the same path,  $-z > 0$ , so we replace that factor by  $(-z)^{1-C}$  and work with the real-axis pair

$$U(z) = {}_2F_1(A, B; C; z), \quad (27a)$$

$$V(z) = (-z)^{1-C} {}_2F_1(A - C + 1, B - C + 1; 2 - C; z). \quad (27b)$$

Here  $U$  is exactly  $\Phi_1$ , while  $V$  plays the role of  $\Phi_2$  in a form that stays real for real  $C$  whenever  $-z > 0$ . Thus  $U$  and  $V$  are two independent real solutions of the hypergeometric ODE on the negative axis where  $z_2$  lives. Any solution  $\Phi$  is a mix of them,

$$\Phi(z) = c_1 U(z) + c_2 V(z),$$

with two coefficients  $c_1, c_2$ . Integer  $C$  does not change that strategy; it only changes how one writes the second basis function (a logarithm may appear), as discussed in [appendix B](#).

Physics fixes only the mix of the two solutions, not the overall scale of  $\Phi$ . From [equation \(26\)](#),  $\hat{S}$  depends on  $\Phi$  only through  $\Phi'/\Phi$ . Scaling both coefficients by the same nonzero factor,

$$(c_1, c_2) \mapsto (\lambda c_1, \lambda c_2),$$

leaves  $\Phi'/\Phi$  unchanged, and therefore leaves  $\hat{S}$  unchanged. The overall scale of  $\Phi$  is free: we never need absolute values of both coefficients. What we do need is the right *mix*, equivalently the ratio  $c_2/c_1$ .

#### 4.3.3 Initial condition and the constant $L$

That mix is fixed by the single initial condition  $\hat{S}(0) = 1$ . At  $\hat{t} = 0$  we have  $z_2 = z_{2,0}$ . Inserting  $\hat{S}(0) = 1$  into [equation \(26\)](#) gives

$$1 = h - \hat{q}_2 z_{2,0} \frac{\Phi'(z_{2,0})}{\Phi(z_{2,0})},$$

and therefore

$$\frac{\Phi'(z_{2,0})}{\Phi(z_{2,0})} = L, \quad L = \frac{h-1}{\hat{q}_2 z_{2,0}}. \quad (28)$$

There is only one physical condition on  $\Phi$ : its logarithmic derivative at the starting point  $z_{2,0}$  must equal the number  $L$ . There is not a second condition that pins down  $c_1$  and  $c_2$  separately.

#### 4.3.4 Choosing the mix: coefficients $K_U$ and $K_V$

It remains only to pick one concrete mix that satisfies [equation \(28\)](#). We write the same linear combination with names  $K_U, K_V$  instead of  $c_1, c_2$ :

$$\Phi(z) = K_U U(z) + K_V V(z). \quad (29)$$

These are simply a particular choice of  $(c_1, c_2)$ , not new unknowns. Write

$$U_0 = U(z_{2,0}), \quad V_0 = V(z_{2,0}), \quad U'_0 = U'(z_{2,0}), \quad V'_0 = V'(z_{2,0})$$

for the basis values at the initial coordinate. We take a convenient Wronskian normalisation,

$$K_U = V'_0 - L V_0, \quad (30a)$$

$$K_V = L U_0 - U'_0. \quad (30b)$$

This is a standard trick for a linear second-order equation. Once a fundamental pair is fixed, the two coefficients above are chosen so that  $\Phi'/\Phi = L$  at  $z_{2,0}$ ; multiplying both by the same nonzero constant does not change  $\Phi'/\Phi$ , so the overall scale of  $(K_U, K_V)$  is free. Only the origin of  $L$  is model-specific. At that point one has

$$\Phi(z_{2,0}) = U_0 V'_0 - U'_0 V_0, \quad \Phi'(z_{2,0}) = L(U_0 V'_0 - U'_0 V_0).$$

The common factor is the Wronskian of  $(U, V)$  at  $z_{2,0}$ , which is nonzero for  $C \notin \mathbb{Z}$ . Hence  $\Phi(z_{2,0}) \neq 0$  and  $\Phi'(z_{2,0})/\Phi(z_{2,0}) = L$  holds at once. Any other multiple  $(\lambda K_U, \lambda K_V)$  would work equally well for  $\hat{S}$ ; the formulae above just pick one scale for us, without hand-solving for a ratio.

The same mix may equally be written  $\Phi = U + D V$  with a single ratio

$$D = \frac{K_V}{K_U} = \frac{L U_0 - U'_0}{V'_0 - L V_0},$$

whenever  $K_U \neq 0$ . In the present probability regime that is automatic: a pure- $V$  solution would tend, as  $z \rightarrow 0^-$ , to a value strictly larger than one, contradicting  $\hat{S} \leq 1$  from [proposition 2.2](#).

#### 4.3.5 Main formula

**Theorem 4.1** (Exact finite-time no-catastrophe probability). *Assume  $\lambda_1 > 0$ ,  $\hat{\lambda}_2 > 0$ ,  $\hat{\delta}_2 > 0$ ,  $\hat{\nu} > 0$ , and  $C \notin \mathbb{Z}$ . With the quantities of this section defined as above, then*

$$\boxed{\hat{S}(\hat{t}) = h - \hat{q}_2 z_2(\hat{t}) \frac{K_U U'(z_2(\hat{t})) + K_V V'(z_2(\hat{t}))}{K_U U(z_2(\hat{t})) + K_V V(z_2(\hat{t}))}} \quad (31)$$

solves [equation \(7a\)](#) with  $\hat{S}(0) = 1$ . Together with [equation \(13\)](#), it gives both finite-time no-catastrophe probabilities. In physical time, after every hatted rate is replaced by the ratio in [equation \(6\)](#),

$$\begin{aligned} S(t) &= h - \hat{q}_2 z_2(\lambda_1 t) \frac{K_U U'(z_2(\lambda_1 t)) + K_V V'(z_2(\lambda_1 t))}{K_U U(z_2(\lambda_1 t)) + K_V V(z_2(\lambda_1 t))}, \\ G(t) &= r_{2,-} - \alpha_2 \frac{z_2(\lambda_1 t)}{1 - z_2(\lambda_1 t)}. \end{aligned} \quad (32)$$

A compact nested evaluation map for every constant appears later in this section ([remark 4.3](#)).

As  $\hat{t} \rightarrow \infty$ , one has  $z_2 \rightarrow 0$  and therefore  $\hat{G} \rightarrow r_{2,-}$  and  $\hat{S} \rightarrow h$ . These limits are the probabilities of avoiding catastrophe forever from one initial type-2 or type-1 individual, respectively. Checks of the initial slopes and degenerate parameter boundaries are collected in [appendix C](#).

The same result can be written as a logarithmic derivative,

$$\hat{S}(\hat{t}) = h - \hat{q}_2 z_2(\hat{t}) \left. \frac{d}{dz} \log \Phi(z) \right|_{z=z_2(\hat{t})}.$$

This compact form exposes the Riccati structure.

*Remark 4.2* (A one-basis special case). If the initial condition happens to give  $K_V = 0$ , equivalently  $D = 0$ , the theorem reduces to

$$\hat{S}(\hat{t}) = h - \hat{q}_2 z_2(\hat{t}) \frac{AB {}_2F_1(A+1, B+1; C+1; z_2(\hat{t}))}{C {}_2F_1(A, B; C; z_2(\hat{t}))}.$$

[Figures 8](#) and [9](#) illustrate the closed formulae for a single rate vector: the no-catastrophe probabilities  $S$  and  $G$ , and the corresponding defective catastrophe-time densities  $-S'$  and  $-G'$ . The solid curves are the exact expressions of [theorem 4.1](#); open markers are an independent numerical check of the backward ODEs by classical fourth-order Runge–Kutta (RK4).

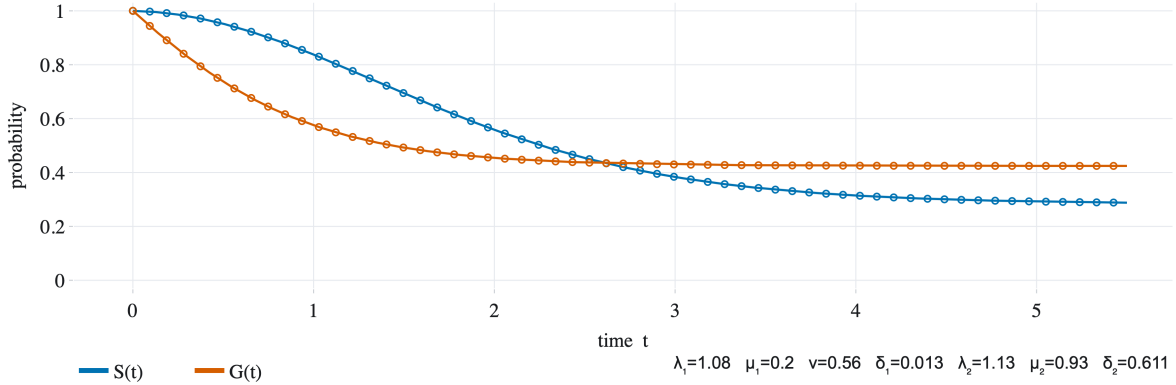


Figure 8: No-catastrophe probabilities  $S(t)$  (type 1 start) and  $G(t)$  (type 2 start) at fixed rates  $\lambda_1 = 1.08$ ,  $\mu_1 = 0.2$ ,  $\nu = 0.56$ ,  $\delta_1 = 0.013$ ,  $\lambda_2 = 1.13$ ,  $\mu_2 = 0.93$ ,  $\delta_2 = 0.611$ . Asymptotic levels  $h = \lim S$  and  $r_{2,-} = \lim G$  are indicated on the panel. Solid curves: closed form of [theorem 4.1](#); open circles: RK4 integration of the unscaled backward ODEs. Rates are labelled on the panel. [Survival visualiser](#)

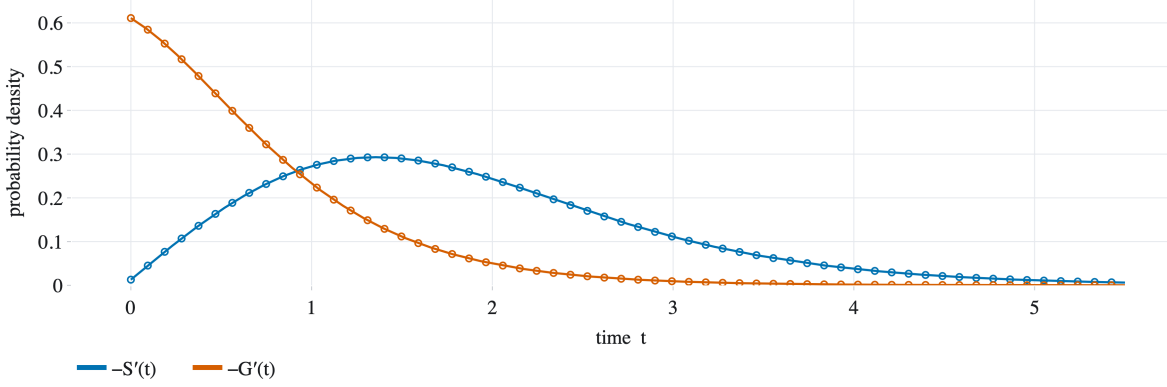


Figure 9: Defective catastrophe-time densities  $-S'(t)$  and  $-G'(t)$  for the same rates as [figure 8](#). Solid curves from the closed form; open circles from the same RK4 check of the backward ODEs.

*Remark 4.3* (Nested parameters). Here every constant in [equations \(31\)](#) and [\(32\)](#) is fixed by the rates as follows. The real hypergeometric basis on the path  $z_2 < 0$  is

$$U(z) = {}_2F_1(A, B; C; z),$$

$$V(z) = (-z)^{1-C} {}_2F_1(A - C + 1, B - C + 1; 2 - C; z).$$

The Wronskian mix enforcing  $\hat{S}(0) = 1$  is then

$$K_U = V'_0 - LV_0, \quad K_V = LU_0 - U'_0,$$

where  $U_0 = U(z_{2,0})$ ,  $V_0 = V(z_{2,0})$ ,  $U'_0 = U'(z_{2,0})$ ,  $V'_0 = V'(z_{2,0})$ , and  $L = (h - 1)/(\hat{q}_2 z_{2,0})$  is the required logarithmic derivative of  $\Phi$  at  $z_{2,0}$ . The hypergeometric parameters are

$$A, B = \frac{C - 1}{2} \pm \sqrt{\frac{(C - 1)^2}{4} - \frac{\hat{\nu}\alpha_2}{\hat{q}_2^2}}, \quad C = 1 + \frac{\hat{\kappa}_1 - 2h}{\hat{q}_2}.$$

The long-time root and type-1 coefficient are

$$h = \frac{\hat{\kappa}_1 - \sqrt{\hat{\kappa}_1^2 - 4(\hat{\mu}_1 + \hat{\nu}r_{2,-})}}{2}, \quad \hat{\kappa}_1 = 1 + \hat{\mu}_1 + \hat{\nu} + \hat{\delta}_1.$$

The type-2 coordinate and its ingredients are

$$z_2(\lambda_1 t) = z_{2,0} e^{\hat{q}_2 \lambda_1 t}, \quad z_{2,0} = \frac{r_{2,-} - 1}{r_{2,+} - 1}, \quad \hat{q}_2 = -\hat{\Delta}_2, \quad \alpha_2 = \frac{\hat{\Delta}_2}{\hat{\lambda}_2},$$

$$r_{2,\pm} = \frac{\hat{\lambda}_2 + \hat{\mu}_2 + \hat{\delta}_2 \pm \hat{\Delta}_2}{2\hat{\lambda}_2}, \quad \hat{\Delta}_2 = \sqrt{(\hat{\lambda}_2 + \hat{\mu}_2 + \hat{\delta}_2)^2 - 4\hat{\lambda}_2\hat{\mu}_2}.$$

Finally the hatted rates are the original rates scaled by the type-1 birth rate,

$$\hat{\mu}_i = \frac{\mu_i}{\lambda_1}, \quad \hat{\nu} = \frac{\nu}{\lambda_1}, \quad \hat{\delta}_i = \frac{\delta_i}{\lambda_1}, \quad \hat{\lambda}_2 = \frac{\lambda_2}{\lambda_1} \quad (i = 1, 2),$$

so the formulae are completely determined by  $(\lambda_1, \mu_1, \nu, \delta_1, \lambda_2, \mu_2, \delta_2)$  and the time  $t$ . Explicit derivatives of  $U$  and  $V$  are recorded in [appendix B](#).

## 4.4 Mathematical interpretation

The formula in [equation \(31\)](#) is exact within the model: it is the finite-time no-catastrophe probability and, by [proposition 2.2](#), the occupation-time average of  $\exp(-\delta_1 \int X - \delta_2 \int Y)$ . The two catastrophe rates do not enter symmetrically. The type-2 rate  $\hat{\delta}_2$  reshapes the roots  $r_{2,\pm}$ , the coordinate  $z_2$ , and the whole function  $\hat{G}$ , which then drives the type-1 equation; the type-1 rate  $\hat{\delta}_1$  enters only through  $\hat{\kappa}_1$ , so it changes  $h$  and the hypergeometric parameters  $A, B, C$  without feeding back into  $\hat{G}$ . That asymmetry is forced by one-way conversion.

We next focus on one potential application of the model: early and adapted intracellular phenotypes of *Y. pestis*, with catastrophe as irreversible loss of host-cell containment.

## 5 Intracellular application: comparing catastrophe-rate regimes

The formula of [section 4](#) holds for any non-negative pair of type specific catastrophe rates  $(\delta_1, \delta_2)$ , and the two rates enter it asymmetrically:  $\delta_2$  reshapes the autonomous type-2 coordinate that drives the whole system, while  $\delta_1$  does not feed back into  $G$ . That asymmetry is not only algebraic. In physical applications, there may be cases where two types lead to a global absorbing event at differing rates. In the case of *Yersinia pestis*, it is scientifically unsettled whether an early or a later bacterial state is more associated with lysis of their host cells. We will explore that application in this section by fixing a reading of the model's components, and use the exact expression to compare four orderings of  $(\delta_1, \delta_2)$ . The aim is not to determine which case is likely to represent physical reality, but to show what an exact finite-time probability makes visible within the context of this modelling regime.

### 5.1 Intracellular mapping and scope

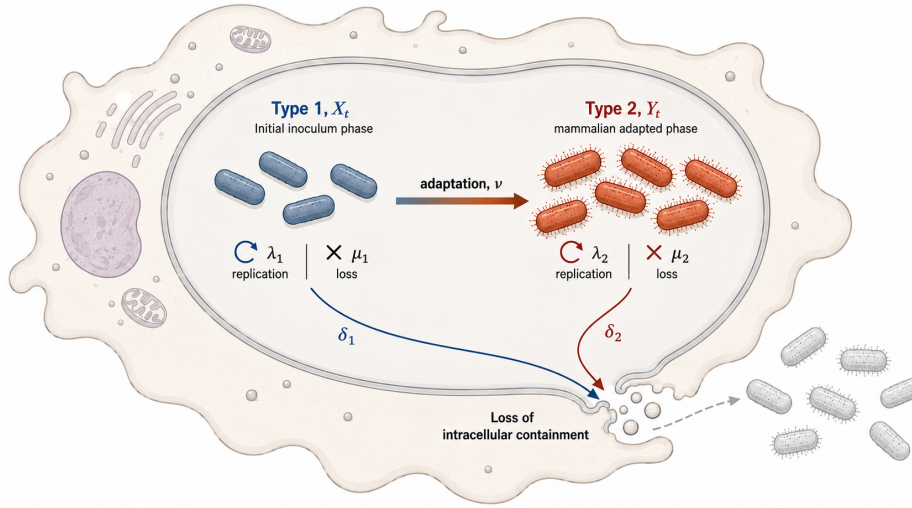
Throughout this section the unit of analysis is a single host cell with a resident bacterial population. Catastrophe means irreversible loss of intracellular containment (burst, lysis, terminal death with release, or another failure of the compartment). It is distinct from ordinary bacterial death of individual cells, which is already encoded by the per-capita rates  $\mu_i$ . Lung-level spread, immune recruitment, transmission, and the extracellular phase after  $\tau_c$  lie outside the process.

Table 1: Reading of mathematical components in an early intracellular infection setting.

| Model component                  | Biological meaning under this reading                 |
|----------------------------------|-------------------------------------------------------|
| $X_t$ (type 1)                   | Early or incompletely adapted intracellular phenotype |
| $Y_t$ (type 2)                   | Later mammalian-adapted intracellular phenotype       |
| Births ( $\lambda_i$ )           | Intracellular replication of that phenotype           |
| Deaths ( $\mu_i$ )               | Ordinary bacterial killing or non-replicative loss    |
| Conversion ( $\nu$ )             | Progressive physiological adaptation (one-way)        |
| Catastrophe rates ( $\delta_i$ ) | Per-capita association with containment failure       |
| Absorbing state                  | Irreversible loss of intracellular containment        |
| Termination at $\tau_c$          | End of the intracellular process for this cell        |

[Table 1](#) is the map used below. The coefficients  $\delta_i$  are effective per-capita *associations* with containment failure under a linear additive intensity  $\delta_1 X_t + \delta_2 Y_t$ ; they are not molecular claims that every adapted bacterium has completed effector injection.

A short biological rationale is enough to make the map usable. In mammalian hosts *Y. pestis* typically begins with an intracellular phase in which a subset of the initial inoculum is taken up by, and survives within, macrophages. Extracellular proliferation follows only after a period of intracellular adaptation. During a time within the relatively protected niche, the bacterium remodels its surface proteins under host temperature and phagosomal stress [4, 5, 6]. Newly internalised bacteria occupy a specialised vacuole, survive and replicate, and reprogramme toward the temperature-induced virulence package of F1 capsule, pH 6 antigen (Psa), and the Ysc–Yop type III secretion system that defines a later anti-phagocytic bacterial phase. Once that programme is in place, adapted bacteria leave the intracellular niche and grow freely outside the cell, resisting re-uptake. The model coarse-grains this continuous intracellular maturation into two states linked by one-way conversion: type 1 represents the early, incompletely adapted bacterial phase; type 2 the later conditioned bacterial phase; conversion  $\nu$  the arming step between them; and catastrophe the irreversible loss of host-cell containment, the release event that ends the single-cell process and opens extracellular outgrowth.



How the intracellular phase ends remains incompletely resolved: host-cell death programmes and timing depend on strain, host background, and infection conditions [7, 8]. Pathogenic yersiniae can kill macrophages; the best-characterised genetically supported link is apoptosis driven by the type III secretion system (T3SS) effector YopJ [9], and components of the temperature-induced virulence package associated with a later anti-phagocytic phenotype may raise the per-capita association with containment failure after intracellular conditioning [8, 10]. That supplies a working hypothesis for a maturation-associated ordering of  $(\delta_1, \delta_2)$ , not a theorem of active type-2-driven rupture. Survival and replication within macrophages can, moreover, be established without treating the T3SS as the main intracellular survival engine [4, 10]. Holding  $(\lambda_i, \mu_i, \nu)$  fixed while comparing  $(\delta_1, \delta_2)$  is therefore a controlled mathematical comparison that is compatible with that literature, rather than a claim of full biological independence between growth and catastrophe.

## 5.2 Four catastrophe-rate regimes

Unequal  $(\delta_1, \delta_2)$  encode different ideas about whether adaptation advances, delays, or leaves unchanged the end of the intracellular phase. We label four regimes for reference. When they are compared later they share the same growth and conversion rates.

- **EQ.**  $\delta_1 = \delta_2 = \delta > 0$ : phenotype-blind per-capita rate (null: failure tracks total load rather than adapted-state machinery).
- **MAT.**  $0 < \delta_1 < \delta_2$ : higher adapted-type rate (maturation-associated). Working hypothesis under the reading above: adaptation raises failure association while residual early risk remains nonzero.
- **GATE.**  $\delta_1 = 0 < \delta_2$ : idealised type-2-only extreme, so failure is possible only after conversion and  $S'(0) = 0$ . Pure GATE may overstate exclusivity of type-2 contribution, because efficient Yop injection is often studied for extracellular bacteria contacting the plasma membrane, whereas bacteria sealed in an intact vacuole may not have the same access.
- **EARLY.**  $\delta_1 > \delta_2 \geq 0$  (illustrations take  $\delta_2 > 0$ ): higher early-type rate, so adaptation may prolong containment (reverse ordering).

Which ordering holds is an empirical question. Under the illustrative normalisation  $\delta_1 + \delta_2 = c$  used in [figure 10](#), GATE sits at  $(0, c)$  while MAT uses  $(\delta_1, c - \delta_1)$  with  $\delta_1 > 0$ ; both coefficients differ, so GATE is the type-2-only boundary of that family with  $S'(0) = 0$ , not simply “MAT with  $\delta_1$  set to zero”.

### 5.3 Exact comparison and limits of inference

From one early founder,  $S(t) = \mathbb{P}_{(1,0)}\{\tau_c > t\}$  is the probability that containment still holds at time  $t$ . Its time derivative supplies the probability density of the catastrophe time:  $f(t) = -S'(t)$  is the unconditional rate at which still-open infections fail at age  $t$ , and  $\int_0^\infty f = 1 - h$  whenever a positive mass  $h = \lim S$  never fails. Among infections that remain contained at  $t$ , the relevant object is not  $f$  itself but the conditional intensity

$$q_S(t) = \frac{f(t)}{S(t)} = -\frac{S'(t)}{S(t)}.$$

Conditioning on  $\{\tau_c > t\}$  discards paths that have already failed, so  $q_S(t) \Delta t$  is the short-time probability of loss of containment in  $[t, t + \Delta t]$  given that the host cell is still intact. The identity  $S'(0) = -\delta_1$  (equivalently  $q_S(0) = \delta_1$ ) is general mathematics: at the instant of founding there is no selection among survivors yet, and the initial hazard is simply the type-1 catastrophe rate.

[Figure 10](#) compares EQ, MAT, GATE, and EARLY under shared growth rates and the coefficient normalisation  $\delta_1 + \delta_2 = c$ . The parameters are illustrative, not fitted. Realised process intensity remains  $\delta_1 X_t + \delta_2 Y_t$ , so equalising the coefficient sum does not equalise realised exposure. Under this normalisation the survival curves  $S(t)$  separate only modestly, whereas the early hazard separates sharply because  $q_S(0) = \delta_1$ , and GATE’s zero intercept is forced by  $S'(0) = 0$ . Later features of both panels are specific to the chosen rates. By one-way conversion,  $G$  depends on  $\delta_2$  alone, so an adapted-founder experiment isolates the type-2 channel in a way that  $S$  cannot. Changing  $\nu$  can advance or shorten engagement of the higher-rate class depending on the ordering, so faster adaptation is not universally more hazardous; elevated risk may also reflect greater type-2 abundance through net growth rather than a larger intrinsic  $\delta_2$ .

The measurements that help most are joint and time-resolved intracellular burden, a phenotype or adaptation marker, and host-cell fate, including the fraction of cells that clear without containment failure. Failure times alone are likely under-identifying: there are trade-offs among  $(\lambda_i, \mu_i, \nu, \delta_i)$ , and survival curves need not separate intrinsic  $\delta_2$  from type-2 abundance. Even under MAT or GATE, agreement with data would support an association between a latent adapted state and

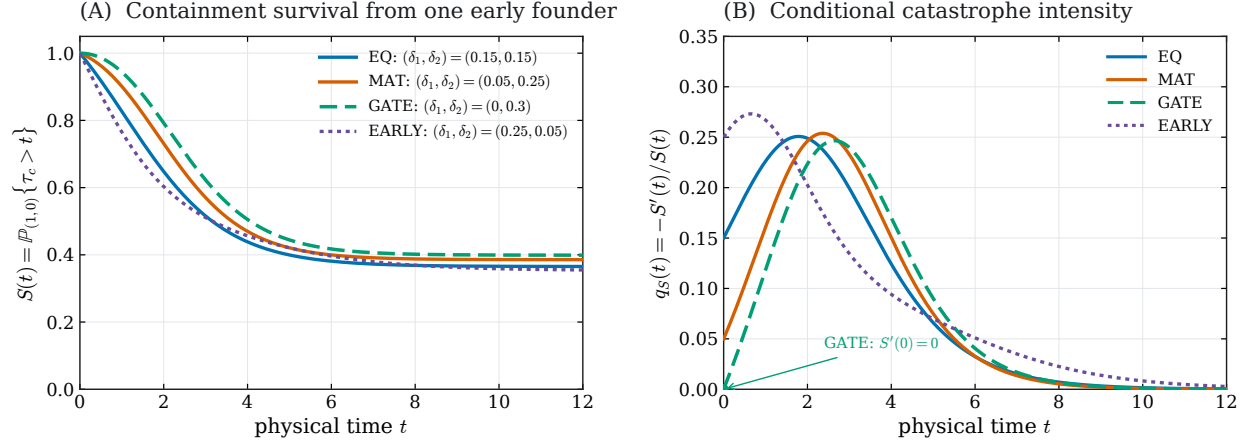


Figure 10: Competing catastrophe-rate regimes under an intracellular *Y. pestis* reading. (A) Survival  $S(t)$  from one early founder; (B) conditional intensity  $q_S = -S'/S$ . Shared growth rates with  $\delta_1 + \delta_2 = c = 0.30$  (coefficient normalisation, not matched realised exposure): EQ (0.15, 0.15), MAT (0.05, 0.25), GATE (0, 0.30), EARLY (0.25, 0.05) ( $\lambda_1 = 1$ ,  $\mu_1 = 0.45$ ,  $\nu = 0.35$ ,  $\lambda_2 = 0.85$ ,  $\mu_2 = 0.40$ ). Parameters are illustrative, not fitted. MAT is a working hypothesis; GATE is the type-2-only extreme ( $S'(0) = 0$ ). Exact closed form for  $S$ ;  $q_S$  from the ODE on the solution path.

failure risk under the stated assumptions, not a proof that adapted bacteria actively rupture the host cell. Alternatives include infection-age markers correlated with burden, host-driven deterioration concurrent with maturation, or mechanisms outside a two-type description. Beyond containment failure itself, the release distributions of [appendix E.4](#) and [figure 13](#) record the pre-catastrophe load and offer a natural next link from these exact laws toward host-level outcomes.

## 6 Conclusion

We set out the finite-time no-catastrophe probability for a two-type population in which either type may trigger one global absorbing event. Because the intensity depends on the whole path, the natural object is an occupation-time transform of the two type abundances rather than a terminal-count generating function alone ([section 2](#)).

The calculation is exact because the backward system is triangular. Solving for  $G$ , shifting  $S$  by its long-time root, linearising the Riccati equation, and changing variable to  $z_2$  produces the Gauss hypergeometric equation and the boxed formula [equation \(31\)](#). The type-2 catastrophe rate reshapes the autonomous coordinate seen by the whole system; the type-1 rate does not feed back into  $G$ . The limits  $h$  and  $r_{2,-}$  are forever-avoidance probabilities. Equal rates and zero-birth boundaries are recorded in [appendix C](#); the equation-level map to Antal and Krapivsky is [appendix D](#).

One natural reading is the early intracellular phase of *Yersinia pestis*: early and adapted phenotypes, with catastrophe as irreversible loss of host-cell containment. The regimes of [section 5](#) then compare orderings of  $(\delta_1, \delta_2)$ .

Beyond the work of this paper, one might consider the release distributions ([appendix E.4](#) and [figure 13](#)) more closely and their implications for a broad infection-success probability within the host. Analysis along these lines may be capable of contributing a quantitative justification for a rate hypothesis favouring MAT/GATE in which the second phase of *Y. pestis* actively triggers burst of the containing cell.

## A From product of survival factors to the pathwise exponential

Along a fixed path the catastrophe clock has instantaneous rate  $\lambda(s)$ . Over a short bin of length  $\Delta t$  the chance of no trigger is about  $1 - \lambda(s)\Delta t$ . Partitioning  $[0, t]$  into  $n$  equal bins of width  $t/n$ , the chance of no trigger on the whole interval is a product of  $n$  such factors. In the constant-rate case the elementary limit

$$\left(1 - \frac{\lambda t}{n}\right)^n \longrightarrow e^{-\lambda t} \quad (n \rightarrow \infty)$$

turns that product into an exponential. With a slowly varying rate the same mesh argument yields

$$\exp\left\{-\int_0^t \lambda(s) ds\right\} = \exp\left\{-\int_0^t (\delta_1 X_s + \delta_2 Y_s) ds\right\},$$

which is the pathwise weight averaged in [proposition 2.2](#).

## B Hypergeometric derivatives and resonant parameters

The derivative of the first basis function follows from the standard identity

$$\frac{d}{dz} {}_2F_1(A, B; C; z) = \frac{AB}{C} {}_2F_1(A+1, B+1; C+1; z).$$

Thus

$$U'(z) = \frac{AB}{C} {}_2F_1(A+1, B+1; C+1; z). \quad (33)$$

For the second basis function, set

$$A_2 = A - C + 1, \quad B_2 = B - C + 1, \quad C_2 = 2 - C.$$

The product rule gives

$$V'(z) = (-z)^{1-C} \left[ \frac{1-C}{z} {}_2F_1(A_2, B_2; C_2; z) + \frac{A_2 B_2}{C_2} {}_2F_1(A_2+1, B_2+1; C_2+1; z) \right]. \quad (34)$$

Equations (33) and (34) show that the main result uses only shifted Gauss hypergeometric functions and elementary factors.

When  $C \in \mathbb{Z}$ , the pair in [equation \(27\)](#) is not generally a fundamental pair because the two Frobenius exponents differ by an integer. A logarithmic second solution can be constructed in the usual way, or the physical solution can be obtained as a continuous limit from nearby noninteger  $C$ . The probability itself is fixed by the regular initial-value problem [equation \(7\)](#); only this convenient basis changes. In computation, direct ODE integration near resonance is a useful independent check.

## C Checks, limits, and the earlier branching model

This appendix records checks and degenerate cases omitted from the main text. We return to the probability first, then to several parameter limits, because a special-function expression is of limited use if its boundaries are unclear. Some limits are smooth inside [equation \(31\)](#); others are better solved before the hypergeometric coordinate is introduced.

## C.1 Initial and infinite-time behaviour

At  $\hat{t} = 0$ , the Wronskian construction in [equations \(29\) and \(30\)](#) forces the logarithmic derivative in [equation \(31\)](#) to equal  $L$ . Consequently  $\hat{S}(0) = 1$ . Differentiation of the backward equations then gives the scaled initial slopes

$$\hat{S}'(0) = -\hat{\delta}_1, \quad \hat{G}'(0) = -\hat{\delta}_2,$$

in agreement with [equation \(4\)](#).

As  $\hat{t}$  grows,  $z_2(\hat{t})$  tends to zero and  $\hat{G}$  tends to  $r_{2,-}$ . The occupation-time representation shows that both probabilities are monotone and hence have limits. Passing to the limit in the type-1 equation with  $\hat{S}' \rightarrow 0$  forces  $\hat{S}$  to a root of [equation \(14\)](#). The upper root exceeds one, while the physical solution is bounded by one, so the long-time value must be the lower root  $h$ . Hence we obtain a useful corollary.

**Corollary C.1** (Avoiding catastrophe forever). *Under the assumptions of [theorem 4.1](#),*

$$\lim_{t \rightarrow \infty} G(t) = r_{2,-}, \quad \lim_{t \rightarrow \infty} S(t) = h.$$

*These are the probabilities that catastrophe never occurs from one initial individual of the corresponding type.*

A birth process can avoid catastrophe forever with positive probability for an ordinary reason: a lineage may die out before accumulating enough exposure to trigger the absorbing event.

## C.2 Equal catastrophe rates

If  $\delta_1 = \delta_2 = \delta$ , then  $\hat{\delta}_1 = \hat{\delta}_2$ . The derivation is unchanged. The common rate enters the type-2 roots through  $\hat{\delta}_2$  and the type-1 stable root through  $\hat{\kappa}_1$ . Thus the unequal-rate result restricts algebraically to the shared-rate model on the diagonal

$$\delta_1 = \delta_2. \tag{35}$$

We do not know of a separate published closed form for the equal-rate case against which a term-by-term expansion could be checked. What can be verified is consistency: under [equation \(35\)](#) the governing equations and every derived parameter reduce as expected.

## C.3 Type 2 cannot trigger catastrophe

Now take  $\delta_2 = 0$ . Starting from one type-2 individual, neither birth nor ordinary death creates catastrophe, and type 2 has no route back to type 1. Therefore

$$G(t) \equiv 1.$$

This is why the root coordinate of [section 3](#) degenerates: the autonomous probability has stopped moving.

In scaled time, the first equation becomes

$$\hat{S}' = \hat{S}^2 - \hat{\kappa}_1 \hat{S} + \hat{\mu}_1 + \hat{\nu}. \tag{36}$$

Let

$$s_{\pm} = \frac{\hat{\kappa}_1 \pm \sqrt{\hat{\kappa}_1^2 - 4(\hat{\mu}_1 + \hat{\nu})}}{2}.$$

If  $\delta_1 > 0$ , then  $s_- < 1 < s_+$  and the roots are distinct. Define

$$R_0 = \frac{1 - s_-}{1 - s_+}. \quad (37)$$

Separation of [equation \(36\)](#) gives

**Proposition C.2** (Elementary type-2-zero boundary). *For  $\delta_2 = 0$  and  $\delta_1 > 0$ ,*

$$\widehat{S}(\hat{t}) = \frac{s_- - R_0 e^{-(s_+ - s_-)\hat{t}} s_+}{1 - R_0 e^{-(s_+ - s_-)\hat{t}}}.$$

*In physical time,*

$$S(t) = \frac{s_- - R_0 e^{-(s_+ - s_-)\lambda_1 t} s_+}{1 - R_0 e^{-(s_+ - s_-)\lambda_1 t}}.$$

One corner of the same family deserves a separate remark. If  $\delta_1 = \delta_2 = 0$ , catastrophe is impossible, so

$$S(t) = G(t) = 1. \quad (38)$$

Here  $1 - s_+$  may vanish in [equation \(37\)](#): the probability itself is regular, but that particular parametrisation is not.

## C.4 No type-1 births

The scaling in [\(5\)](#) excludes  $\lambda_1 = 0$ , although the original process does not. At this boundary the type-1 equation is no longer Riccati. It is linear:

$$S' = -(\mu_1 + \nu + \delta_1)S + \mu_1 + \nu G.$$

An integrating factor therefore gives the exact unscaled solution

$$S(t) = e^{-(\mu_1 + \nu + \delta_1)t} \left[ 1 + \int_0^t e^{(\mu_1 + \nu + \delta_1)s} (\mu_1 + \nu G(s)) ds \right]. \quad (39)$$

Here  $G$  is the elementary solution of [equation \(3b\)](#); it has the root form of [proposition 3.1](#) with unscaled type-2 rates when  $\lambda_2, \delta_2 > 0$ , and the corresponding linear or constant boundary form otherwise. Thus the absence of type-1 births is an exactly solvable degeneration, not a singularity of the probability model.

## C.5 No type-2 births

There is another coordinate boundary when  $\lambda_2 = 0$ . Assume here that  $\lambda_1 > 0$ ,  $\hat{\nu} > 0$ , and  $\hat{\delta}_2 > 0$ ; the zero-conversion and zero catastrophe-rate cases are already elementary. Define

$$k = \hat{\mu}_2 + \hat{\delta}_2, \quad g_\infty = \frac{\hat{\mu}_2}{k}, \quad a = \hat{\nu}(1 - g_\infty) = \frac{\hat{\nu}\hat{\delta}_2}{k}.$$

Then

$$\widehat{G}(\hat{t}) = g_\infty + (1 - g_\infty)e^{-k\hat{t}}. \quad (40)$$

In physical time,  $G(t) = g_\infty + (1 - g_\infty)e^{-k\lambda_1 t}$ . Let  $h_0$  be the lower root of

$$h_0^2 - \hat{\kappa}_1 h_0 + \hat{\mu}_1 + \hat{\nu}g_\infty = 0,$$

and put

$$d_0 = \hat{\kappa}_1 - 2h_0, \quad \rho = \frac{d_0}{k}, \quad w(\hat{t}) = w_0 e^{-k\hat{t}/2}, \quad w_0 = \frac{2\sqrt{a}}{k}.$$

Primes in the next display mean differentiation with respect to  $w$ . With

$$M = \frac{2 - \hat{\kappa}_1}{kw_0}, \quad \begin{aligned} c_J &= Y'_\rho(w_0) - MY_\rho(w_0), \\ c_Y &= MJ_\rho(w_0) - J'_\rho(w_0), \end{aligned} \quad (41)$$

define

$$\Theta(w) = c_J J_\rho(w) + c_Y Y_\rho(w),$$

where  $J_\rho$  and  $Y_\rho$  are the Bessel functions of the first and second kinds.

**Proposition C.3** (Zero type-2 birth boundary). *Under the assumptions of this subsection,*

$$\hat{S}(\hat{t}) = \frac{\hat{\kappa}_1}{2} + \frac{k}{2} w(\hat{t}) \frac{\Theta'(w(\hat{t}))}{\Theta(w(\hat{t}))} \quad (42)$$

solves the type-1 backward equation with  $\hat{S}(0) = 1$ . In physical time,

$$S(t) = \frac{\hat{\kappa}_1}{2} + \frac{k}{2} w(\lambda_1 t) \frac{\Theta'(w(\lambda_1 t))}{\Theta(w(\lambda_1 t))}.$$

Moreover  $\hat{S}(\hat{t}) \rightarrow h_0$  and  $\hat{G}(\hat{t}) \rightarrow g_\infty$ , so  $S(t) \rightarrow h_0$  and  $G(t) \rightarrow g_\infty$ .

*Proof.* Equation (40) follows from the now-linear type-2 equation. Write  $\hat{S} = h_0 + X$  and use  $X = -Z'/Z$ . The type-1 equation becomes

$$Z'' + d_0 Z' + a e^{-k\hat{t}} Z = 0.$$

Set  $Z = e^{-d_0 \hat{t}/2} \Theta(w)$ . The equation for  $\Theta$  is

$$w^2 \Theta'' + w \Theta' + (w^2 - \rho^2) \Theta = 0,$$

which is Bessel's equation. Returning through the substitution gives [equation \(42\)](#). The coefficients in [equation \(41\)](#) enforce  $\Theta'(w_0)/\Theta(w_0) = M$ ; their denominator-free form is valid because the Bessel Wronskian gives  $\Theta(w_0) = 2/(\pi w_0)$ . Finally the bounded probability branch contains the small- $w$  behaviour  $w^{-\rho}$  and hence tends to the lower root  $h_0$ . The other local behaviour would tend to  $\hat{\kappa}_1 - h_0 > 1$ .  $\square$

## C.6 No conversion

When  $\nu = 0$ , the two types do not communicate. Each no-catastrophe probability solves its own one-type Riccati equation. The type-2 solution is already [equation \(13\)](#). For type 1, the same root construction applies to

$$\hat{S}' = \hat{S}^2 - (1 + \hat{\mu}_1 + \hat{\delta}_1) \hat{S} + \hat{\mu}_1.$$

The hypergeometric equation is unnecessary in this case. The boundary can be obtained by continuity from the central formula along non-degenerate parameter paths, but the direct scalar solution is clearer and numerically safer.

## D Equation-level recovery of Antal–Krapivsky

This appendix records the precise relation to Antal and Krapivsky [2]. Their observable is the generating function of the terminal population counts; ours is the probability that a global catastrophe has not occurred. The branching skeleton (two types, one-way conversion, triangular backward structure) is shared (figure 11).

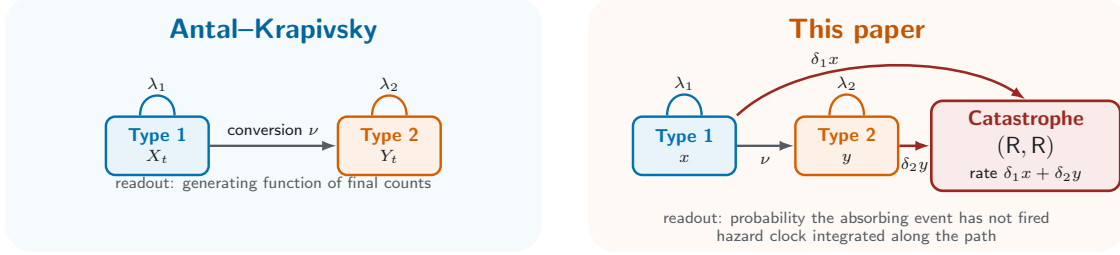


Figure 11: Antal–Krapivsky terminal-population generating functions (left) versus the no-catastrophe probabilities studied here (right). Both use the same two-type one-way branching skeleton. Left:  $\mathcal{A}(x, y, t) = \mathbb{E}[x^{X_t} y^{Y_t}]$  with initial data  $\mathcal{A}(x, y, 0) = x$ ,  $\mathcal{B}(x, y, 0) = y$ , and no global absorbing catastrophe. Right:  $S(t) = \mathbb{P}_{(1,0)}(\tau_c > t)$  and  $G(t) = \mathbb{P}_{(0,1)}(\tau_c > t)$  with  $S(0) = G(0) = 1$  and catastrophe rate  $\delta_1 x + \delta_2 y$ .

To make the equation-level map exact, set  $\delta_1 = \delta_2 = 0$  and let  $\mathcal{A}(x, y, t)$  and  $\mathcal{B}(x, y, t)$  denote the generating functions starting from one type-1 and one type-2 individual. Their initial data are

$$\mathcal{A}(x, y, 0) = x, \quad \mathcal{B}(x, y, 0) = y,$$

rather than the survival values  $S(0) = G(0) = 1$ . The backward system is then

$$\begin{aligned} \partial_t \mathcal{A} &= \lambda_1 \mathcal{A}^2 - (\lambda_1 + \mu_1 + \nu) \mathcal{A} + \mu_1 + \nu \mathcal{B}, \\ \partial_t \mathcal{B} &= \lambda_2 \mathcal{B}^2 - (\lambda_2 + \mu_2) \mathcal{B} + \mu_2. \end{aligned}$$

Under the rate map

$$\alpha_i^{\text{AK}} = \lambda_i, \quad \beta_i^{\text{AK}} = \mu_i, \quad \nu^{\text{AK}} = \nu,$$

these are exactly the triangular equations used in that paper, before its normalisation  $\alpha_1^{\text{AK}} = 1$ .

If instead the catastrophe rates vanish while our survival data are kept, equation (38) follows and both probabilities are identically one. Recovering the earlier governing system therefore requires the generating-function initial data  $x, y$ ; the no-catastrophe formula alone does not yield a nontrivial terminal population distribution.

## E A catastrophe-aware population generating function

We can retain population-count information without removing catastrophe from the model. For this purpose, use a killed, or defective, generating function. Its coefficients record the counts on paths that have not yet reached catastrophe. For  $i = 1, 2$ , define

$$\mathcal{F}_1(x, y, t) = \mathbb{E}_{(1,0)}[x^{X_t} y^{Y_t} \mathbf{1}_{\{\tau_c > t\}}], \quad (43a)$$

$$\mathcal{F}_2(x, y, t) = \mathbb{E}_{(0,1)} \left[ x^{X_t} y^{Y_t} \mathbf{1}_{\{\tau_c > t\}} \right]. \quad (43b)$$

The variable  $x$  plays no role in  $\mathcal{F}_2$ , but we keep both arguments so that the definitions remain parallel. We may also take the expectation over the catastrophe-free branching skeleton and write

$$\mathcal{F}_i(x, y, t) = \mathbb{E}_i \left[ x^{X_t} y^{Y_t} \exp \left\{ - \int_0^t (\delta_1 X_s + \delta_2 Y_s) ds \right\} \right].$$

Hence,

$$[x^m y^n] \mathcal{F}_i(x, y, t) = \mathbb{P}_i \{ X_t = m, Y_t = n, \tau_c > t \},$$

and the total masses give the no-catastrophe probabilities:

$$\mathcal{F}_1(1, 1, t) = S(t), \quad \mathcal{F}_2(1, 1, t) = G(t).$$

Thus these functions are generally subnormalised at  $(x, y) = (1, 1)$ . Their mass deficit is the probability that catastrophe has occurred by time  $t$ .

**Proposition E.1** (Backward system for the killed generating functions). *The functions in [equation \(43\)](#) satisfy*

$$\partial_t \mathcal{F}_1 = \lambda_1 \mathcal{F}_1^2 - (\lambda_1 + \mu_1 + \nu + \delta_1) \mathcal{F}_1 + \mu_1 + \nu \mathcal{F}_2, \quad (44a)$$

$$\partial_t \mathcal{F}_2 = \lambda_2 \mathcal{F}_2^2 - (\lambda_2 + \mu_2 + \delta_2) \mathcal{F}_2 + \mu_2, \quad (44b)$$

with

$$\mathcal{F}_1(x, y, 0) = x, \quad \mathcal{F}_2(x, y, 0) = y. \quad (45)$$

*Proof.* Condition on the first event, as in [section 2](#). A birth produces two independent descendant lineages, which gives  $\mathcal{F}_i^2$ . Ordinary death leaves the empty population and contributes one, while conversion replaces  $\mathcal{F}_1$  by  $\mathcal{F}_2$ . A catastrophe contributes zero. This last point explains why  $\delta_i$  appears in the negative linear coefficient but not in the constant term: the catastrophe rate cannot be absorbed into the ordinary death rate.  $\square$

If  $\delta_1 = \delta_2 = 0$ , [equations \(44\) and \(45\)](#) becomes the Antal–Krapivsky generating-function system recovered in [appendix D](#). With positive catastrophe rates it is instead a Feynman–Kac deformation of that system, and its total mass need not equal one.

## E.1 The absorbing state and conditional populations

To represent absorbed paths explicitly, let  $\rho$  mark the absorbing state R and set

$$\mathcal{H}_i(x, y, \rho, t) = \mathcal{F}_i(x, y, t) + \rho [1 - \mathcal{F}_i(1, 1, t)]. \quad (46)$$

The coefficient of  $\rho$  is the probability of catastrophe by time  $t$ . Hence,

$$\mathcal{H}_i(1, 1, 1, t) = 1.$$

Because the process stops at its first catastrophe, [equation \(46\)](#) is linear in  $\rho$ . After absorption there is no finite pair of population counts.

On paths that avoid catastrophe up to time  $t$ , the population PGF is

$$\mathcal{Q}_i(x, y, t) = \frac{\mathcal{F}_i(x, y, t)}{\mathcal{F}_i(1, 1, t)}. \quad (47)$$

For example,  $\partial_x \mathcal{Q}_i(1, 1, t)$  and  $\partial_y \mathcal{Q}_i(1, 1, t)$  are the conditional mean type-1 and type-2 counts. To record the population just before catastrophe, use the flux generating function

$$\mathcal{J}_i(x, y, t) = \delta_1 x \partial_x \mathcal{F}_i(x, y, t) + \delta_2 y \partial_y \mathcal{F}_i(x, y, t).$$

This follows coefficientwise:

$$[x^m y^n] \mathcal{J}_i = (\delta_1 m + \delta_2 n) \mathbb{P}_i\{X_t = m, Y_t = n, \tau_c > t\},$$

the right-hand side is the time density of catastrophe from the pre-absorption state  $(m, n)$ . Summing over all counts gives

$$\mathcal{J}_1(1, 1, t) = -S'(t), \quad \mathcal{J}_2(1, 1, t) = -G'(t).$$

## E.2 Reuse of the hypergeometric solution

The full killed PGF requires no new differential equation. We first rescale time:

$$\hat{\mathcal{F}}_i(x, y, \hat{t}) = \mathcal{F}_i(x, y, \hat{t}/\lambda_1)$$

and keep the notation  $r_{2,\pm}$ ,  $\alpha_2$ ,  $\hat{q}_2$ ,  $h$ , and  $A, B, C$  from [sections 3 and 4](#). In the generic regime of [theorem 4.1](#), only the starting coordinate changes:

$$z_{2,0}(y) = \frac{y - r_{2,-}}{y - r_{2,+}}, \quad z_2(\hat{t}; y) = z_{2,0}(y) e^{\hat{q}_2 \hat{t}}. \quad (48)$$

Then the autonomous solution is

$$\hat{\mathcal{F}}_2(x, y, \hat{t}) = r_{2,-} - \alpha_2 \frac{z_2(\hat{t}; y)}{1 - z_2(\hat{t}; y)}. \quad (49)$$

Suppose  $z_{2,0}(y) \neq 0$ . Let  $W_1, W_2$  be a fundamental pair of the Gauss equation [\(21\)](#), chosen consistently along the path in [equation \(48\)](#). Write

$$L(x, y) = \frac{h - x}{\hat{q}_2 z_{2,0}(y)}$$

and define

$$\begin{aligned} K_1(x, y) &= W_2'(z_{2,0}(y)) - L(x, y) W_2(z_{2,0}(y)), \\ K_2(x, y) &= L(x, y) W_1(z_{2,0}(y)) - W_1'(z_{2,0}(y)). \end{aligned}$$

Then, with  $z = z_2(\hat{t}; y)$ ,

$$\boxed{\hat{\mathcal{F}}_1(x, y, \hat{t}) = h - \hat{q}_2 z \frac{K_1 W_1'(z) + K_2 W_2'(z)}{K_1 W_1(z) + K_2 W_2(z)}}. \quad (50)$$

The Wronskian calculation in [equation \(30\)](#) ensures that  $\hat{\mathcal{F}}_1(x, y, 0) = x$ . When  $y = 1$ , the trajectory follows the negative real path used in the main theorem, so we may take  $(W_1, W_2) = (U, V)$ . Then [equations \(49\) and \(50\)](#) reduce to  $S$  and  $G$  at  $x = y = 1$ .

The exceptional value  $z_{2,0}(y) = 0$  follows by continuity. It can also be obtained directly because the type-2 solution is then constant. Extracting every coefficient with a bivariate Cauchy integral requires complex  $x$  and  $y$ , so the second hypergeometric solution must use a consistent complex branch. One can avoid this branch choice by integrating [equation \(44\)](#) directly on the contour. Derivatives at  $(x, y) = (1, 1)$ , including the conditional moments, stay on the original real branch.

### E.3 Three distributions encoded by the killed PGF

To keep the formulas readable, write

$$p_{m,n}^{(i)}(t) = [x^m y^n] \mathcal{F}_i(x, y, t), \quad \sigma_i(t) = \mathcal{F}_i(1, 1, t),$$

so that  $\sigma_1 = S$  and  $\sigma_2 = G$ . Normalising these coefficients gives the three distributions below.

**Counts immediately before catastrophe.** Once catastrophe occurs, the process enters the cemetery state R. The counts of interest are therefore the left limits  $(X_{\tau_c-}, Y_{\tau_c-})$ . Define

$$j_{m,n}^{(i)}(t) = (\delta_1 m + \delta_2 n) p_{m,n}^{(i)}(t).$$

From state  $(m, n)$ , catastrophe occurs at rate  $\delta_1 m + \delta_2 n$ . Thus  $j_{m,n}^{(i)}(t)$  is the joint density of the catastrophe time and the state just before it. Conditional on catastrophe at time  $t$ , in the density sense, the PGF is

$$\mathcal{R}_i(x, y \mid t) = \frac{\mathcal{J}_i(x, y, t)}{\mathcal{J}_i(1, 1, t)},$$

and hence

$$\mathbb{P}_i\{X_{\tau_c-} = m, Y_{\tau_c-} = n \mid \tau_c = t\} = \frac{j_{m,n}^{(i)}(t)}{\mathcal{J}_i(1, 1, t)}.$$

If we disregard the exact catastrophe time, the distribution conditional on catastrophe by a horizon  $T$  has PGF

$$\mathcal{R}_{i,T}^{\text{cat}}(x, y) = \frac{\int_0^T \mathcal{J}_i(x, y, t) dt}{1 - \sigma_i(T)}.$$

As  $T \rightarrow \infty$ , this becomes the population distribution at eventual catastrophe, conditional on catastrophe occurring. Integrating the flux over all future times and assigning the forever-avoidance mass to the origin produces the unconditional ultimate release law of [appendix E.4](#).

**Counts given no catastrophe.** At a fixed time  $t$ , [equation \(47\)](#) gives

$$\mathbb{P}_i\{X_t = m, Y_t = n \mid \tau_c > t\} = \frac{p_{m,n}^{(i)}(t)}{\sigma_i(t)}.$$

Ordinarily extinct paths remain in this distribution and contribute mass at  $(0, 0)$ .

**Counts given neither catastrophe nor ordinary fixation.** Here “fixation” means ordinary extinction at  $(0, 0)$ . Its no-catastrophe mass at time  $t$  is

$$e_i(t) = \mathcal{F}_i(0, 0, t).$$

If the process is still active and catastrophe-free, its conditional PGF is

$$\mathcal{Q}_i^{\text{act}}(x, y, t) = \frac{\mathcal{F}_i(x, y, t) - e_i(t)}{\sigma_i(t) - e_i(t)}.$$

Thus, for  $m + n > 0$ ,

$$\mathbb{P}_i\{X_t = m, Y_t = n \mid \tau_c > t, X_t + Y_t > 0\} = \frac{p_{m,n}^{(i)}(t)}{\sigma_i(t) - e_i(t)}.$$

Conditioning on eventual ordinary extinction before catastrophe leaves a point mass at  $(0, 0)$ . Its probability is  $h$  from type 1 or  $r_{2,-}$  from type 2.

Figure 12 displays the three laws at a fixed observational time, started from one type-1 individual. Colour scales are panel-wise (not shared across the row).

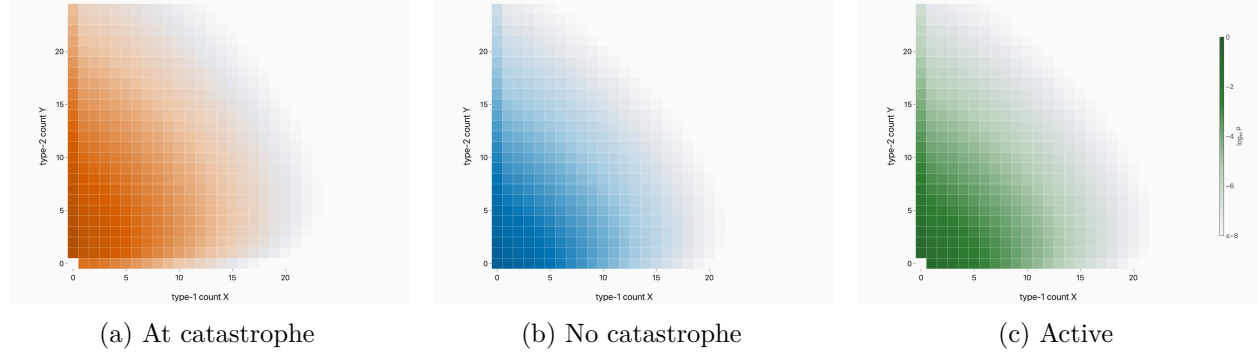


Figure 12: Three distributions encoded by the killed PGF at fixed time  $t = 3$ , from one type-1 individual. (a) Pre-catastrophe counts conditional on catastrophe at time  $t$  (density sense). (b) Counts conditional on no catastrophe up to  $t$  (includes ordinary extinction at the origin). (c) Active counts conditional on no catastrophe and non-extinction (log scale). Rates:  $\lambda_1 = 0.75$ ,  $\mu_1 = 0.3$ ,  $\nu = 0.7$ ,  $\delta_1 = 0.001$ ,  $\lambda_2 = 0.75$ ,  $\mu_2 = 0.25$ ,  $\delta_2 = 0.083$ . Colour intensity scales are independent across panels. [Distribution visualiser](#)

## E.4 Ultimate release counts

The three conditional laws above do not yet record the full unconditional law of the pre-catastrophe state, including paths that never release. Define the ultimate type-1 and type-2 release counts by

$$(X_{\text{rel}}, Y_{\text{rel}}) = \begin{cases} (X_{\tau_c-}, Y_{\tau_c-}) & \text{on } \{\tau_c < \infty\}, \\ (0, 0) & \text{on } \{\tau_c = \infty\}. \end{cases} \quad (51)$$

Thus release is the left-limit population at the absorbing event when that event occurs, and is set to zero when no release event ever takes place.

*Remark E.2* (Generic regime versus boundaries). In the generic regime  $\delta_1 > 0$ ,  $\delta_2 > 0$ , the only way to have  $\tau_c = \infty$  is ordinary extinction to  $(0, 0)$  before the catastrophe clock fires. The origin atom of  $(X_{\text{rel}}, Y_{\text{rel}})$  is then exactly the forever-avoidance probability, and coincides with clearance without release. On certain catastrophe-rate boundaries (for instance  $\delta_1 = 0$ ), paths with  $\tau_c = \infty$  need not be extinct; [equation \(51\)](#) still assigns  $(0, 0)$  on those paths, now meaning “no release event” rather than “empty burst”. The formulae below remain valid on those boundaries; only the biological reading of the origin atom changes.

The joint law follows by integrating the catastrophe flux of [appendix E.3](#). For  $m + n \geq 1$ , the mass at  $(m, n)$  is the integrated density of first entrance to  $(R, R)$  from that pre-absorption state. The complementary forever-avoidance mass sits at the origin. Explicitly, with  $\sigma_i(\infty) = \lim_{t \rightarrow \infty} \sigma_i(t)$  (so  $\sigma_1(\infty) = h$  and  $\sigma_2(\infty) = r_{2,-}$ ),

$$\mathbb{P}_i\{X_{\text{rel}} = m, Y_{\text{rel}} = n\} = \begin{cases} \sigma_i(\infty), & (m, n) = (0, 0), \\ \int_0^\infty (\delta_1 m + \delta_2 n) p_{m,n}^{(i)}(t) dt, & m + n \geq 1. \end{cases} \quad (52)$$

These probabilities sum to one. The positive-support contribution equals  $\int_0^\infty \mathcal{J}_i(1, 1, t) dt = \int_0^\infty (-\sigma'_i(t)) dt = 1 - \sigma_i(\infty)$ , since  $\sigma_i(0) = 1$  and  $\sigma_i \downarrow \sigma_i(\infty)$ .

The corresponding joint probability generating function is therefore

$$\boxed{\begin{aligned} \mathcal{P}_i^{\text{rel}}(x, y) &:= \mathbb{E}_i[x^{X_{\text{rel}}} y^{Y_{\text{rel}}}] \\ &= \sigma_i(\infty) + \int_0^\infty \mathcal{J}_i(x, y, t) dt. \end{aligned}} \quad (53)$$

Coefficient extraction recovers [equation \(52\)](#):  $\mathbb{P}_i\{X_{\text{rel}} = m, Y_{\text{rel}} = n\} = [x^m y^n] \mathcal{P}_i^{\text{rel}}(x, y)$ .

When  $\sigma_i(\infty) < 1$ , the law conditional on eventual catastrophe is supported on  $\{(m, n) : m+n \geq 1\}$  and has generating function

$$\mathcal{R}_i^{\text{cat}}(x, y) := \mathbb{E}_i[x^{X_{\text{rel}}} y^{Y_{\text{rel}}} \mid \tau_c < \infty] = \frac{\int_0^\infty \mathcal{J}_i(x, y, t) dt}{1 - \sigma_i(\infty)}. \quad (54)$$

This is the  $T \rightarrow \infty$  limit of the horizon object  $\mathcal{R}_{i,T}^{\text{cat}}$  from [appendix E.3](#). The unconditional release PGF is the two-point mixture

$$\mathcal{P}_i^{\text{rel}}(x, y) = \sigma_i(\infty) \cdot 1 + (1 - \sigma_i(\infty)) \mathcal{R}_i^{\text{cat}}(x, y). \quad (55)$$

For numerical evaluation, the joint masses in [equation \(52\)](#) may be obtained by integrating the killed master equations on a truncated lattice and accumulating the catastrophe flux; [figure 13](#) shows one such computation.

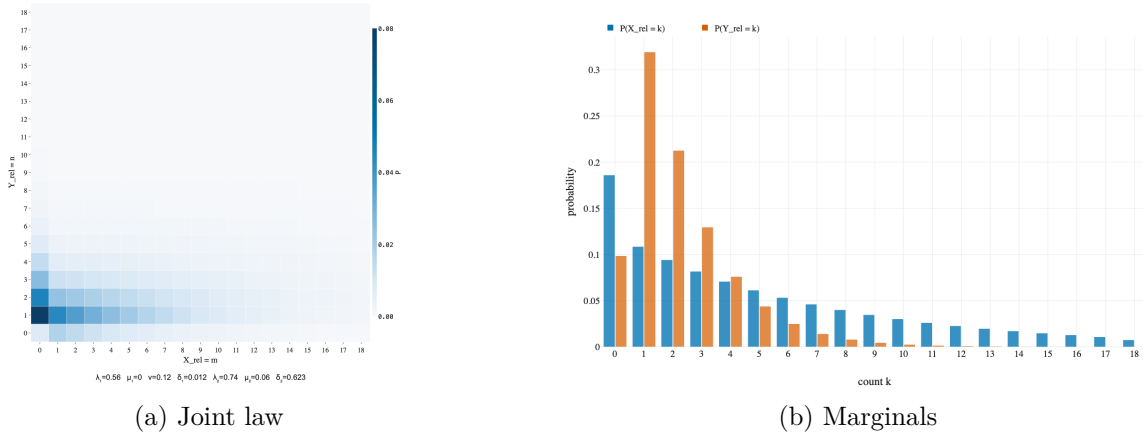


Figure 13: Unconditional ultimate release law  $(X_{\text{rel}}, Y_{\text{rel}})$  from one type-1 individual. (a) Joint masses  $\mathbb{P}_{(1,0)}\{X_{\text{rel}} = m, Y_{\text{rel}} = n\}$ ; the origin atom is forever-avoidance  $\sigma_1(\infty) = h$  (clearance without release in the generic regime). (b) Marginals  $\mathbb{P}\{X_{\text{rel}} = k\}$  (blue) and  $\mathbb{P}\{Y_{\text{rel}} = k\}$  (vermillion). Rates (on panel (a)):  $\lambda_1 = 0.56$ ,  $\mu_1 = 0$ ,  $\nu = 0.12$ ,  $\delta_1 = 0.012$ ,  $\lambda_2 = 0.74$ ,  $\mu_2 = 0.06$ ,  $\delta_2 = 0.623$ . Lattice truncation of the killed master equations; residual active mass at the numerical horizon is negligible. [Release visualiser](#)

The marginal generating functions are the specialisations  $\mathcal{P}_i^{\text{rel}}(x, 1)$  and  $\mathcal{P}_i^{\text{rel}}(1, y)$ . For  $m \geq 1$ ,

$$\mathbb{P}_i\{X_{\text{rel}} = m\} = \int_0^\infty \sum_{n=0}^\infty (\delta_1 m + \delta_2 n) p_{m,n}^{(i)}(t) dt, \quad (56)$$

and for  $n \geq 1$ ,

$$\mathbb{P}_i\{Y_{\text{rel}} = n\} = \int_0^\infty \sum_{m=0}^\infty (\delta_1 m + \delta_2 n) p_{m,n}^{(i)}(t) dt. \quad (57)$$

Both marginals carry at least the origin atom  $\sigma_i(\infty)$ , but the event  $\{X_{\text{rel}} = 0\}$  is strictly larger than  $\{X_{\text{rel}} = Y_{\text{rel}} = 0\}$  whenever catastrophe can occur from a pure type-2 state. Thus  $\mathbb{P}_i\{X_{\text{rel}} = 0\} \geq \sigma_i(\infty)$ , with equality if and only if pure type-2 states cannot trigger (in particular if  $\delta_2 = 0$ ). The same remark applies to  $Y_{\text{rel}}$  with the roles of the types reversed.

Unconditional means  $\mathbb{E}_i[X_{\text{rel}}]$  and  $\mathbb{E}_i[Y_{\text{rel}}]$  are the first partial derivatives of  $\mathcal{P}_i^{\text{rel}}$  at  $(1, 1)$ ; conditional means given eventual catastrophe follow by dividing by  $1 - \sigma_i(\infty)$ .

## References

- [1] Krishna B. Athreya and Peter E. Ney. *Branching Processes*. Springer–Verlag, Berlin, 1972.
- [2] Tibor Antal and P. L. Krapivsky. Exact solution of a two-type branching process: Models of tumor progression. *Journal of Statistical Mechanics: Theory and Experiment*, 2011(08):P08018, 2011. <https://doi.org/10.1088/1742-5468/2011/08/P08018>.
- [3] Carl M. Bender and Steven A. Orszag. *Advanced Mathematical Methods for Scientists and Engineers*. McGraw–Hill, New York, 1978.
- [4] Céline Pujol and James B. Bliska. The ability to replicate in macrophages is conserved between *Yersinia pestis* and *Yersinia pseudotuberculosis*. *Infection and Immunity*, 71(10):5892–5899, 2003.
- [5] Michael G. Connor, Amanda R. Pulsifer, Donghoon Chung, Eric C. Rouchka, Brian K. Ceresa, and Matthew B. Lawrenz. *Yersinia pestis* targets the host endosome recycling pathway during the biogenesis of the *Yersinia*-containing vacuole to avoid killing by macrophages. *mBio*, 9(1):e01800–17, 2018.
- [6] Yuehua Ke, Zeliang Chen, and Ruifu Yang. *Yersinia pestis*: Mechanisms of entry into and resistance to the host cell. *Frontiers in Cellular and Infection Microbiology*, 3:106, 2013.
- [7] Ayelet Zauberman, Sara Cohen, Emanuelle Mamroud, Yehuda Flashner, Avital Tidhar, Raphael Ber, Eitan Elhanany, Avigdor Shafferman, and Baruch Velan. Interaction of *Yersinia pestis* with macrophages: Limitations in YopJ-dependent apoptosis. *Infection and Immunity*, 74(6):3239–3250, 2006.
- [8] Kristen N. Peters, Miqdad O. Dhariwala, Jennifer M. Hughes Hanks, Corrie R. Brown, and Deborah M. Anderson. Early apoptosis of macrophages modulated by injection of *Yersinia pestis* YopK promotes progression of primary pneumonic plague. *PLoS Pathogens*, 9(4):e1003324, 2013.
- [9] Denise M. Monack, Joan Mecsas, Nafisa Ghori, and Stanley Falkow. *Yersinia* signals macrophages to undergo apoptosis and YopJ is necessary for this cell death. *Proceedings of the National Academy of Sciences of the United States of America*, 94(19):10385–10390, 1997.
- [10] Céline Pujol and James B. Bliska. Turning *Yersinia* pathogenesis outside in: Subversion of macrophage function by intracellular yersiniae. *Clinical Immunology*, 114(3):216–226, 2005.